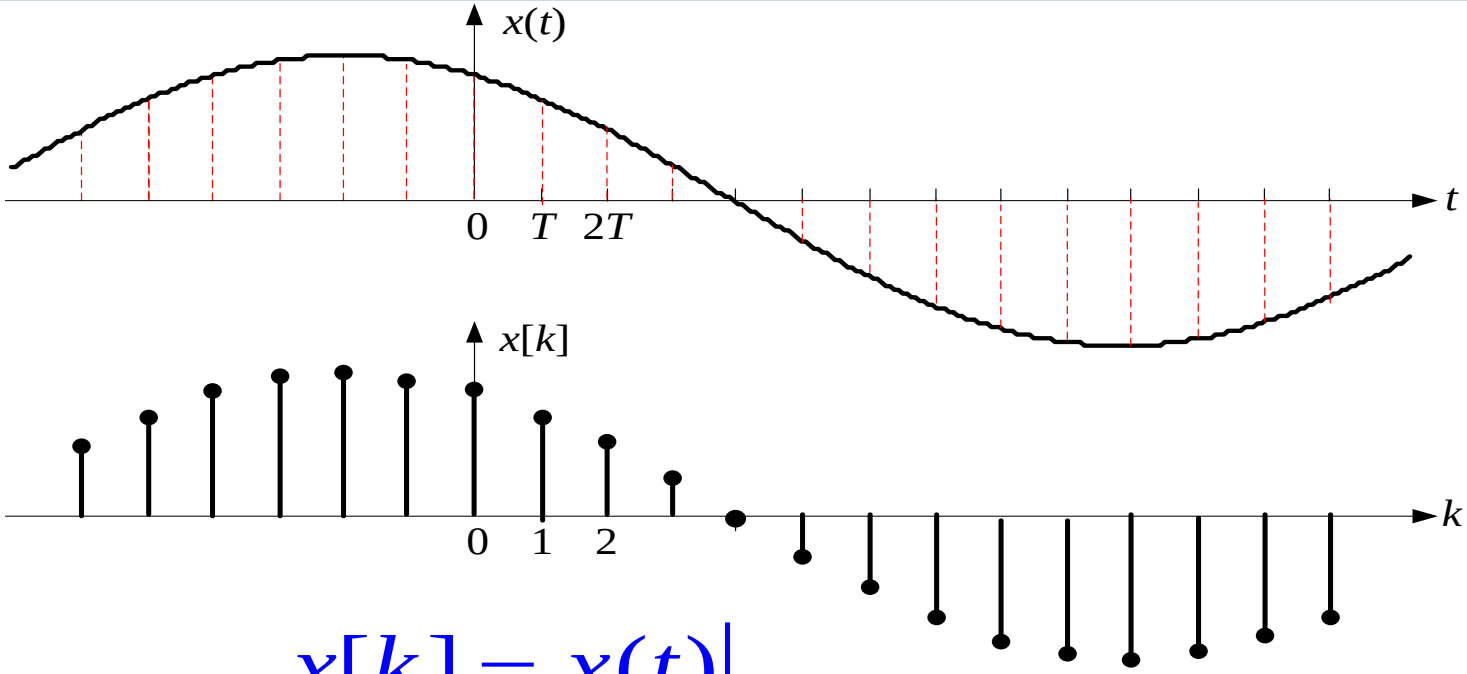
Digital Signal Processing

Zhou Hang

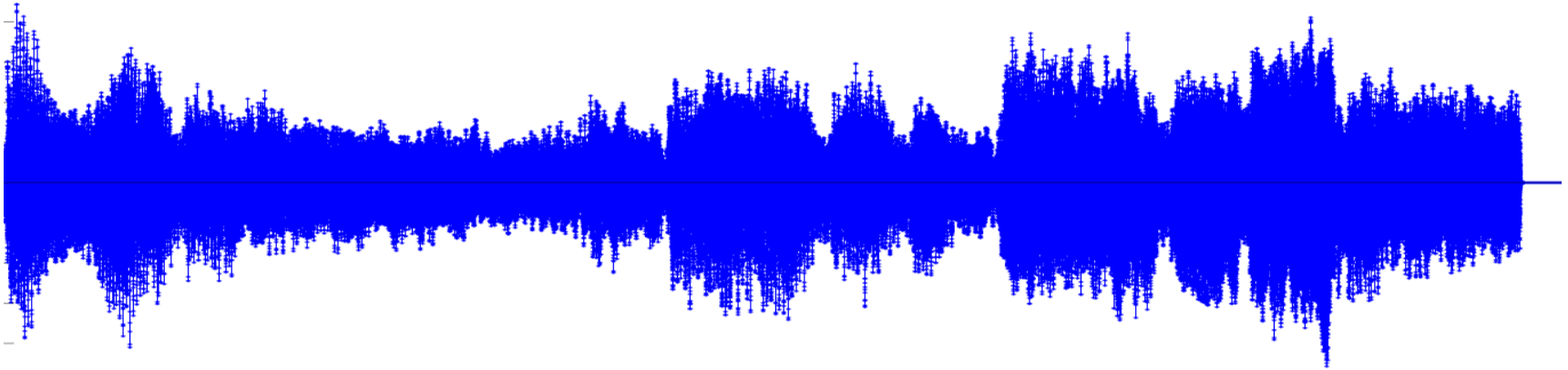
- What is signal sampling
- Why conduct sampling
- How to perform signal sampling
- Analysis of Signal Sampling Theory
- Application of Sampling Theorem in Engineering
- Reconstruction of sampled signals
- Discrete processing of continuous signals

What is signal sampling



$$x[k] = x(t)\big|_{t=kT}$$

What is signal sampling





[x,Fs,Bits]=wavread(′damaige′)

play(x)

Fs=44,100; Bits=16

What is signal sampling



Who proposed signal sampling



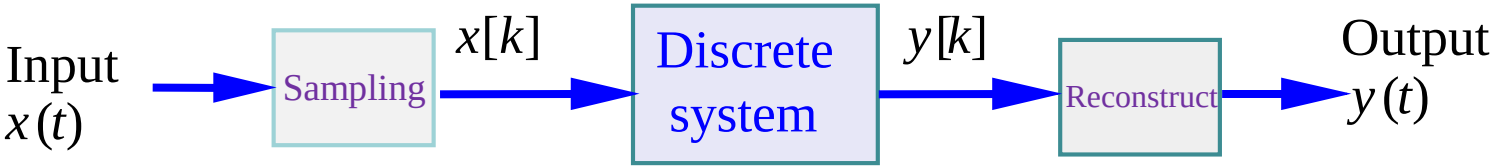
Harry Nyquist(1889 – 1976)

Nyquist， American physicist, born in Sweden in 1889. He made significant contributions to information theory. He immigrated to the United States in 1907 and enrolled in North Dakota University for further studies in 1912. In 1917, he obtained a doctoral degree in physics from Yale University. From 1917 to 1934, he worked at AT&T and later transferred to Bell Telephone Laboratories.

In 1927, Nyquist determined that for sampling a finite time continuous signal of a certain bandwidth, in order to avoid "half wave loss" in the original waveform, the sampling rate should be at least twice the highest frequency of the signal. This is the famous Nyquist sampling theorem.

Why conduct signal sampling

Lancaster University College at Beijing Jiaotong University



Using digital technology to process analog signals

The main advantages of discrete signals and systems are:

※ Good signal stability: The data is represented in binary and is less affected by external factors.

※ High signal reliability: lossless storage and anti-interference transmission.

※ Easy signal processing: signal compression, signal encoding, signal encryption, etc.

※ High accuracy of the system: The accuracy of the system can be improved by increasing the word length.

※ Strong system flexibility: Changing the coefficients of the system enables it to perform different functions.

How to perform signal sampling

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The influence of sampling frequency on speech signals

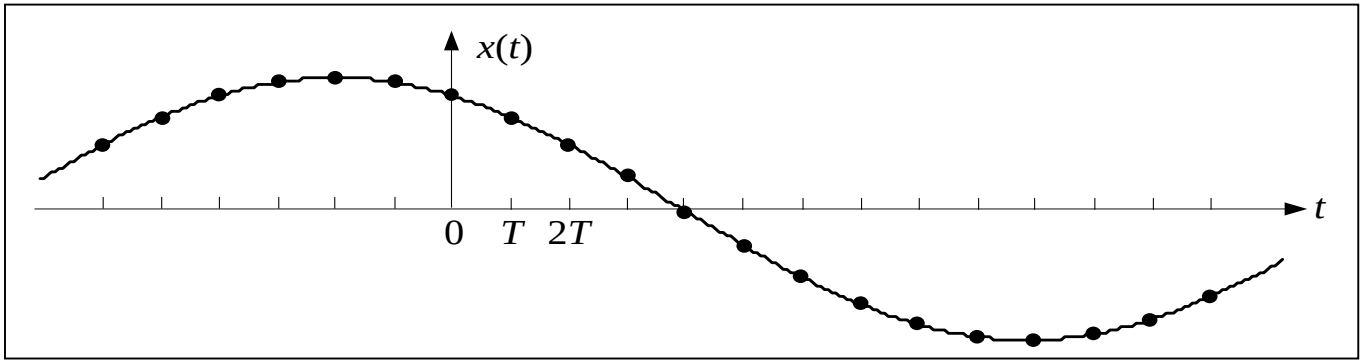
f_{sam}=44,100 Hz

f_{sam}=22,050 Hz

f_{sam}=11,025 Hz

f_{sam}=5,512 Hz

How to perform signal sampling

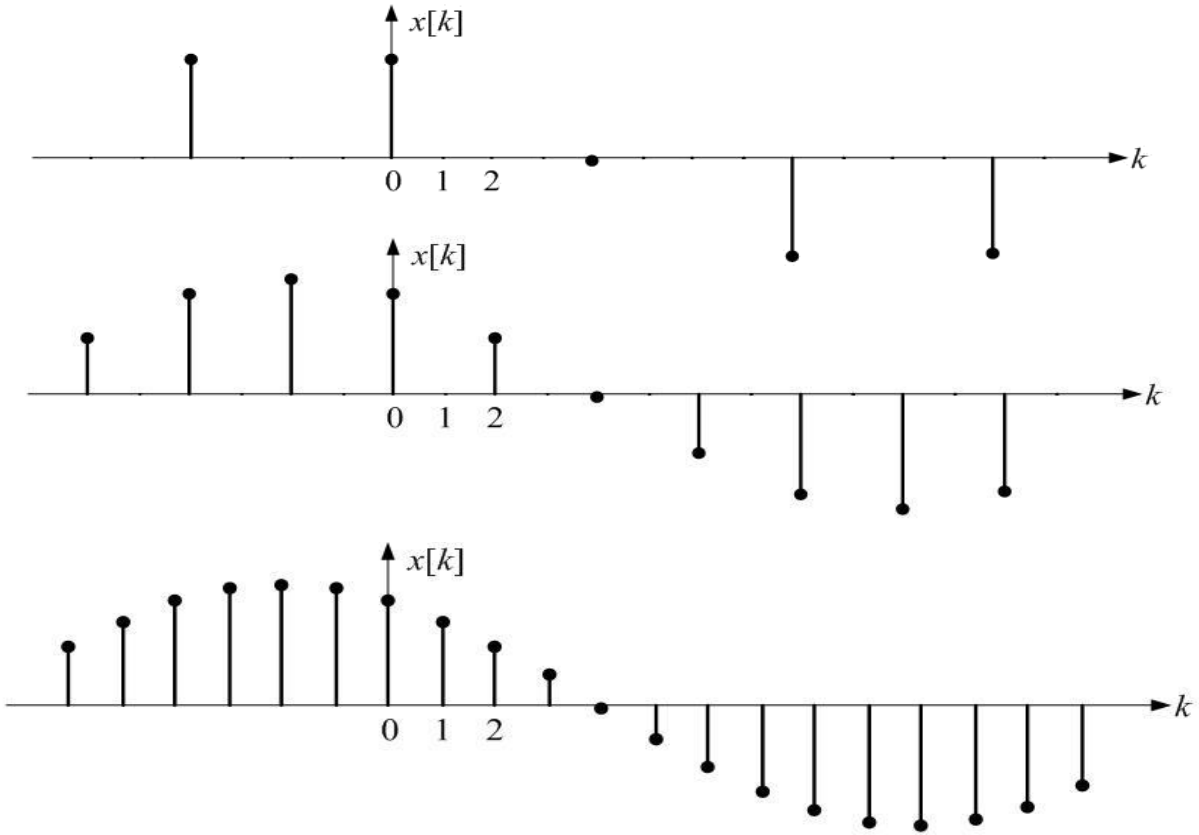


$$x[k] = x(t)\Big|_{t=kT}$$

How to select the sampling interval T?

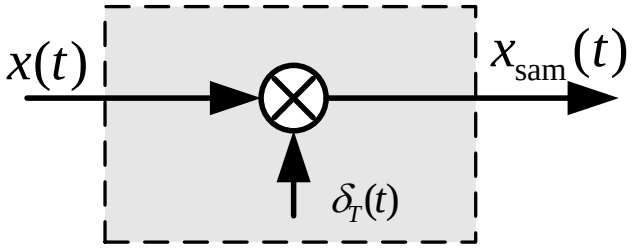
How to perform signal sampling

Lancaster University College at Beijing Jiaotong University



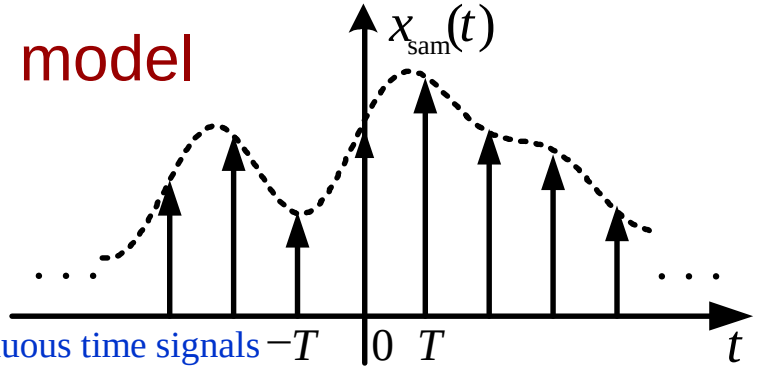
Theoretical analysis of signal time-domain sampling

Lancaster University College at Beijing Jiaotong University

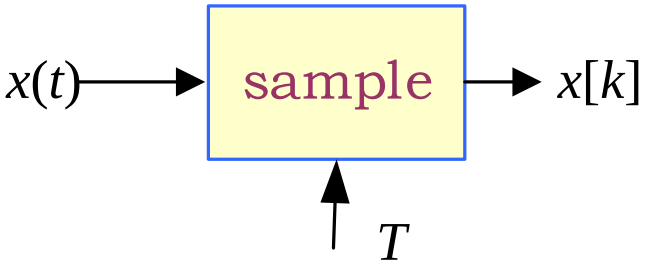


Signal ideal sampling model

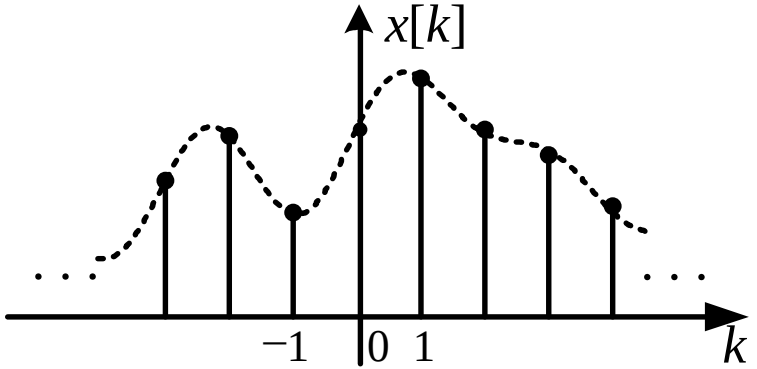
Traditional model



Both input and output are continuous time signals

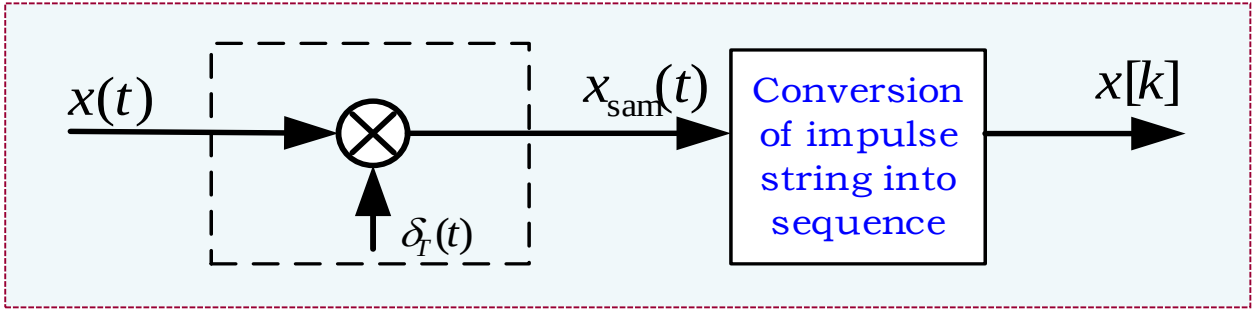


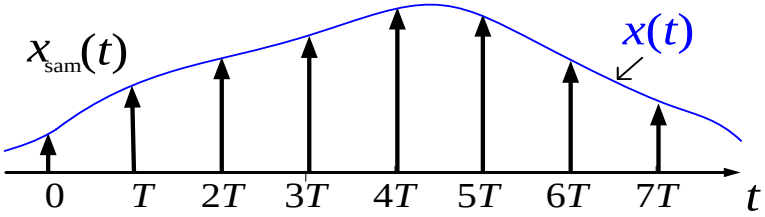
New Model

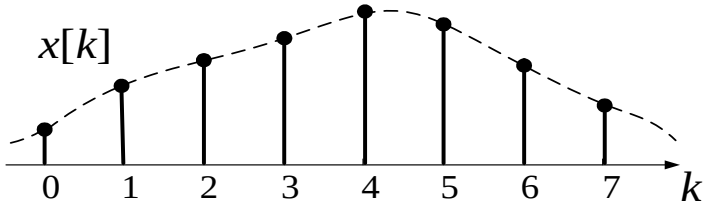


The input is a continuous time signal, and the output is a discrete-time signal

Theoretical analysis of signal time-domain sampling



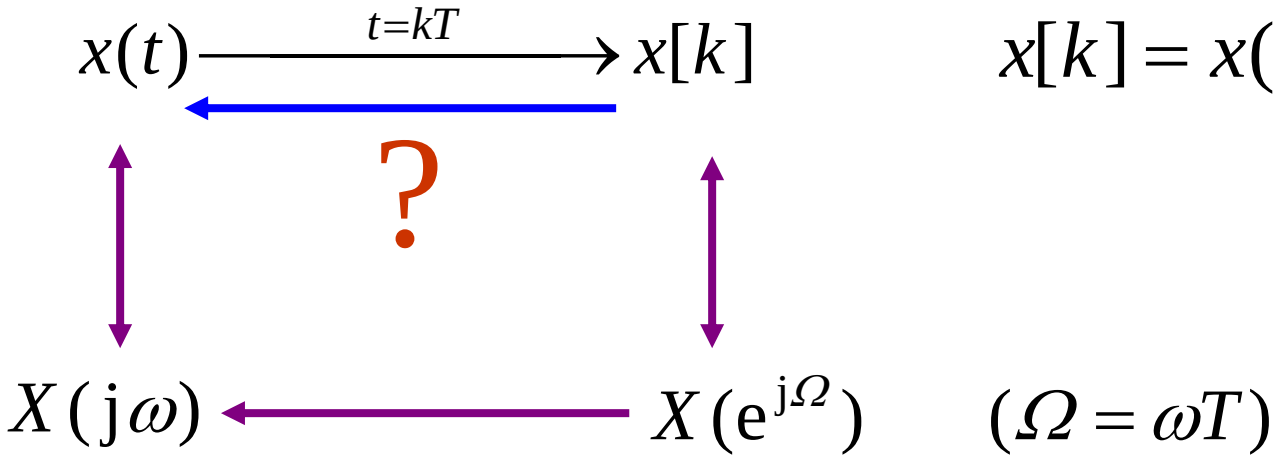




$$x_{\text{sam}}(t) = x(t)\delta_T(t) = \sum_{k=-\infty}^{+\infty} x(kT)\delta(t-kT)$$

$x[k]$

Theoretical analysis of signal time-domain sampling



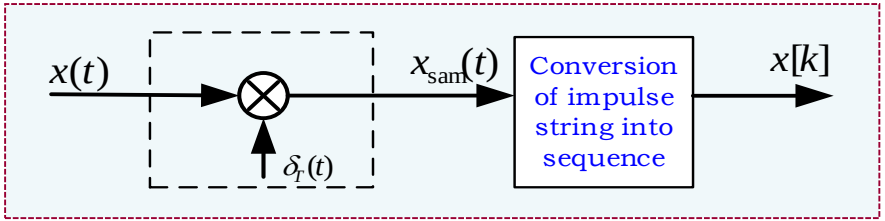
$$x[k] = x(t)\big|_{t=kT}$$

$$(\Omega = \omega T)$$

The frequency spectrum of the continuous signal $x(t)$ is $X(\mathrm{j}w)$,

The frequency spectrum of the discrete sequence $x[k]$ is $X(\mathrm{e}^{\mathrm{j}\Omega})$.

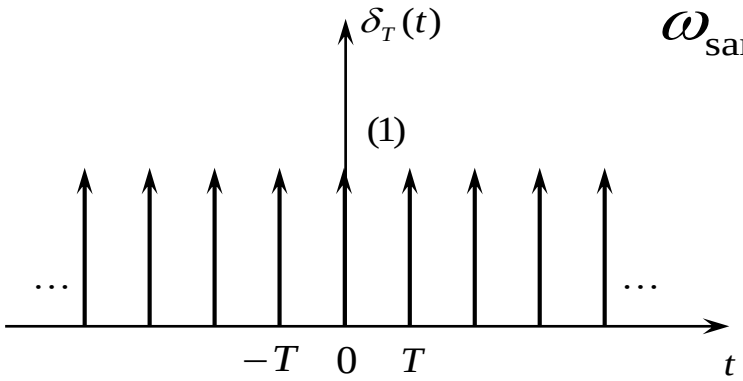
Theoretical analysis of signal time-domain sampling

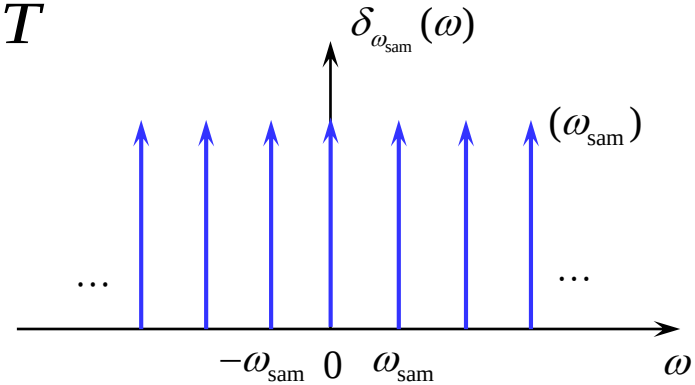


$$\delta_T(t) = \sum_{k=-\infty}^{+\infty} \delta(t - kT)$$

$$\delta_{\omega_{\text{sam}}}(\omega) = \omega_{\text{sam}} \sum_{n=-\infty}^{+\infty} \delta(\omega - n\omega_{\text{sam}})$$

$$\omega_{\text{sam}} = 2\pi/T$$





Theoretical analysis of signal time-domain sampling

$$x_{\text{sam}}(t) = x(t)\delta_T(t) = \sum_{k=-\infty}^{+\infty} x(kT)\delta(t-kT)$$

$$X_{\text{sam}}(\text{j}\omega) = \frac{1}{2\pi} X(\text{j}\omega) * \omega_{\text{sam}} \sum_{n=-\infty}^{+\infty} \delta(\omega - n\omega_{\text{sam}})$$

$$= \frac{1}{T} \sum_{n=-\infty}^{+\infty} X\left[\text{j}(\omega - n\omega_{\text{sam}})\right]$$

$$X_{\text{sam}}(\text{j}\omega) = \sum_{k=-\infty}^{+\infty} x(kT)\text{e}^{-\text{j}k\omega T} = \sum_{k=-\infty}^{+\infty} x[k]\text{e}^{-\text{j}k\Omega} = X(\text{e}^{\text{j}\Omega}) \quad \Omega=\omega T$$

Time domain sampling theorem

If $x[k] = x(t)\big|_{t=kT}$

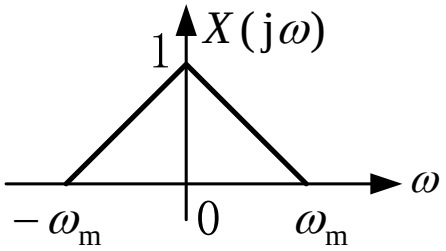
Then $$X(e^{j\Omega}) = \frac{1}{T}\sum_{n=-\infty}^{+\infty} X\left[j(\omega - n\omega_{sam})\right]$$ $\Omega=\omega T$

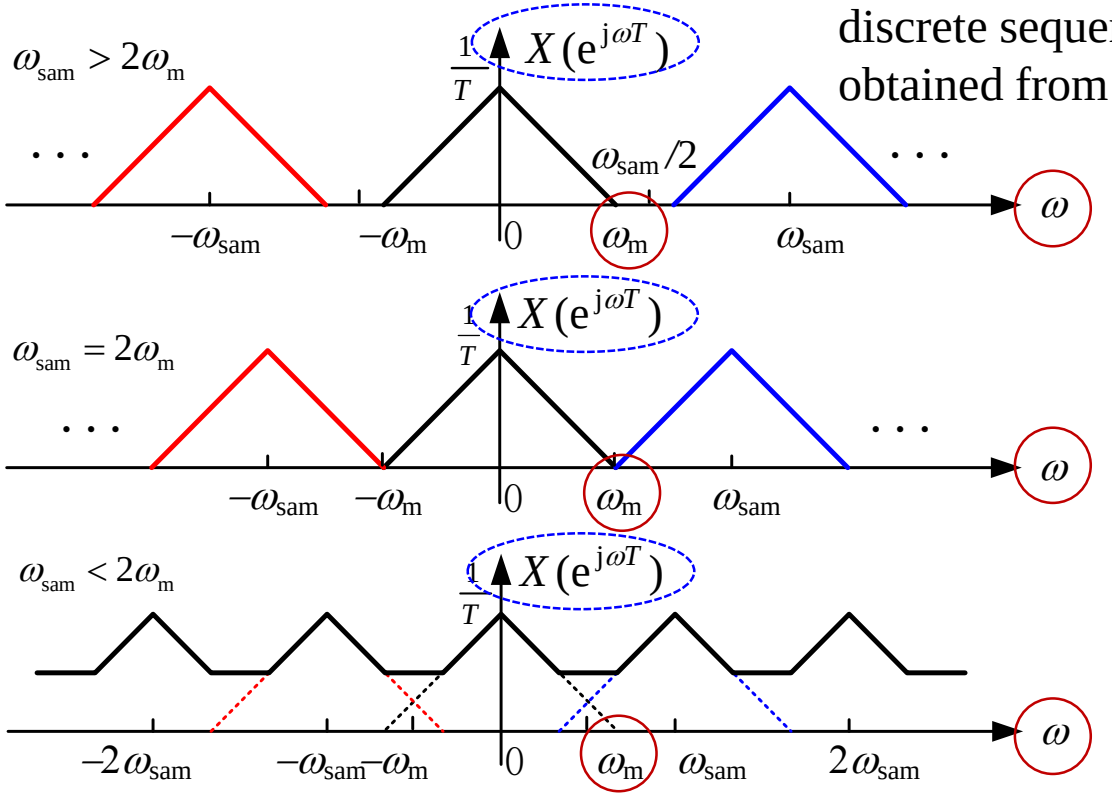
The discretization of the signal in the time domain leads to the periodization of its frequency domain.

Among them, T is the sampling interval, $w_{sam}=2\pi /T$ is the sampling angular frequency.

$$X(e^{j\Omega}) = \frac{1}{T}\sum_{n=-\infty}^{+\infty} X\left[j(\omega - n\omega_{sam})\right] \qquad (\Omega = \omega T)$$

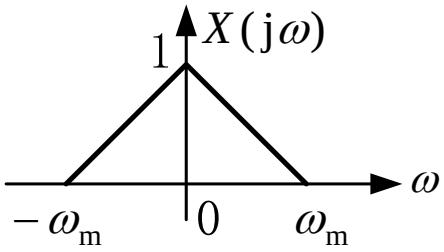
The spectrum of the discrete sequence obtained from sampling

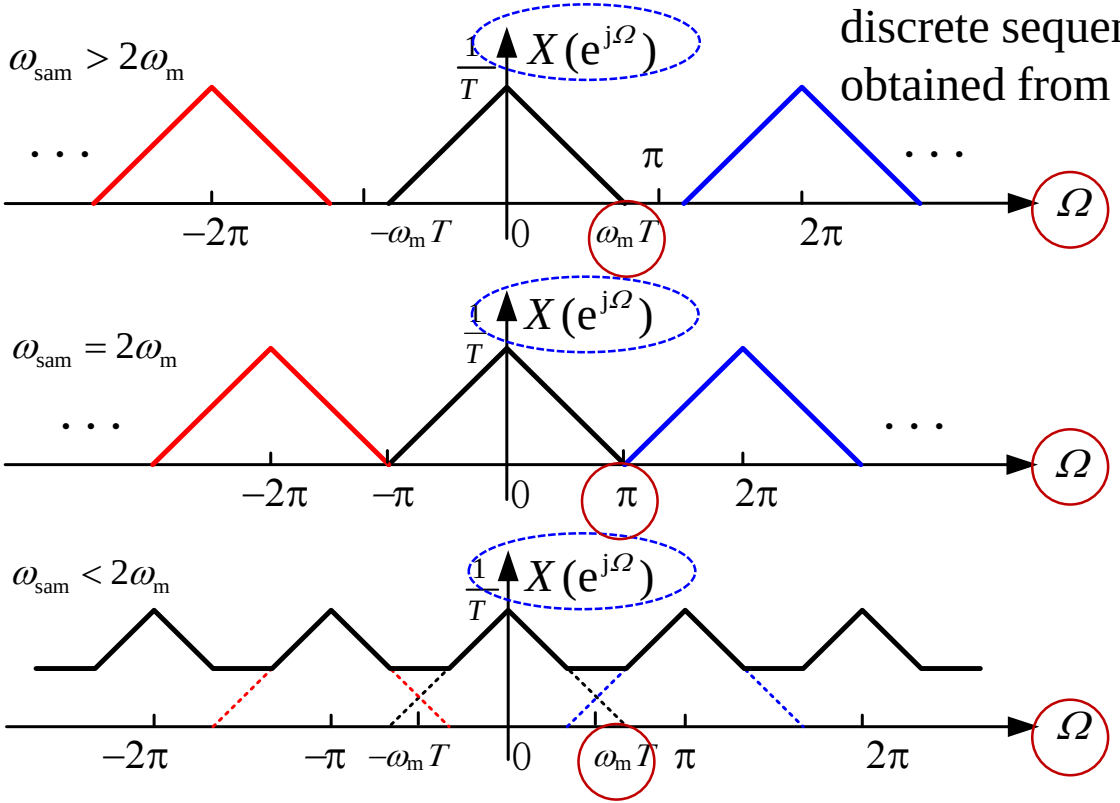




$$X(e^{j\Omega}) = \frac{1}{T}\sum_{n=-\infty}^{+\infty} X\left[j(\omega - n\omega_{sam})\right] \qquad (\Omega = \omega T)$$

The spectrum of the discrete sequence obtained from sampling





Theoretical analysis of signal time-domain sampling

Exception: Band limited signal

If the highest angular frequency of the band limited signal $x(t)$ is w_m, then under certain conditions, the signal $x(t)$ can be uniquely represented by equidistant T samples.

The sampling interval T must meet the following requirements:

$$T \le \pi/\omega_{\mathrm{m}} = 1/(2f_{\mathrm{m}})$$

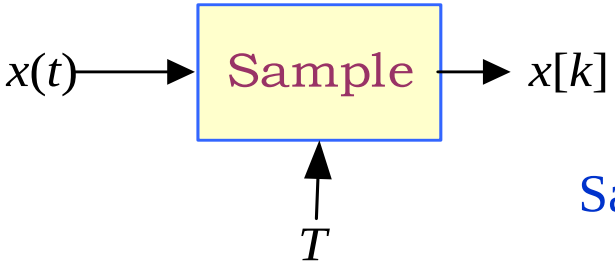
$$f_{\mathrm{sam}} \ge 2f_{\mathrm{m}} \quad (\text{或}\omega_{\mathrm{sam}} \ge 2\omega_{\mathrm{m}})$$

$f_{\mathrm{sam}} = 2f_{\mathrm{m}}$ is the minimum sampling frequency, known as Nyquist Rate.

Application of Sampling Theorem in Engineering

- Implementation of Signal Sampling Engineering
- Sampling of non band limited signals
- Sampling of narrowband high-frequency signals
- The highest frequency of the signal is unknown
- Railway control signal recognition

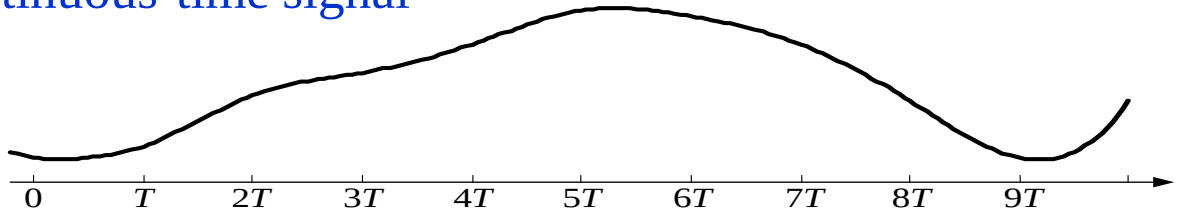
Application of Sampling Theorem in Engineering



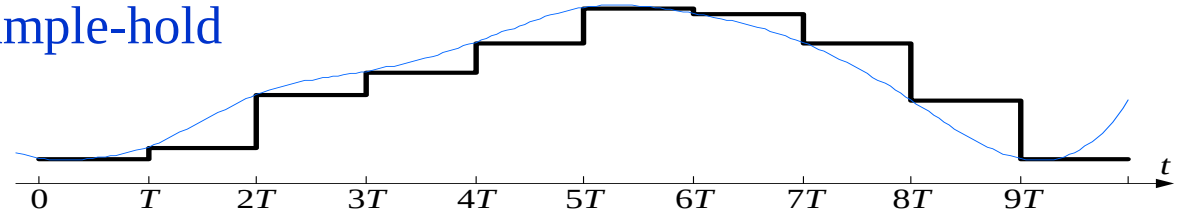


Quantization coding

Continuous-time signal

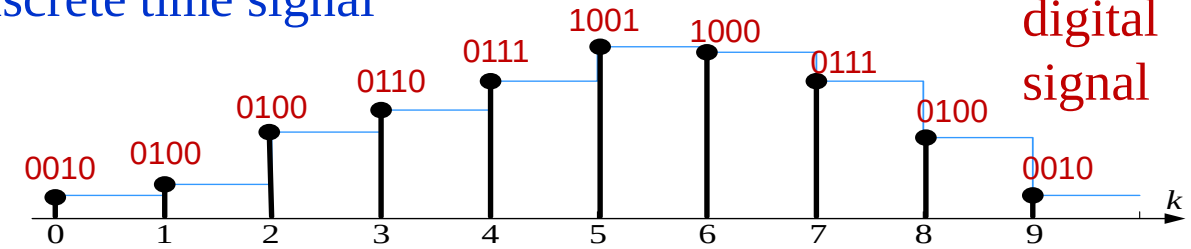


Sample-hold



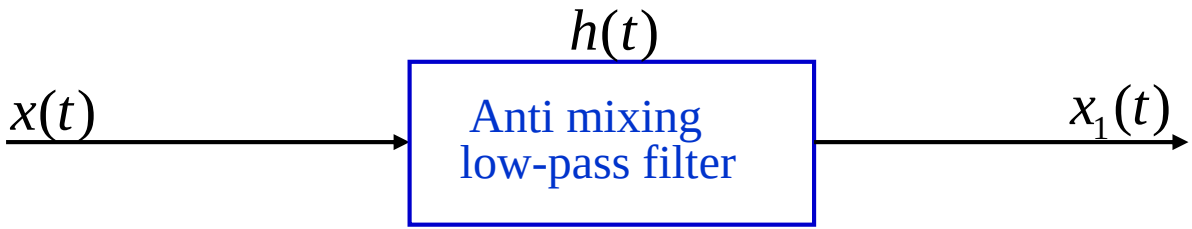
Discrete time signal

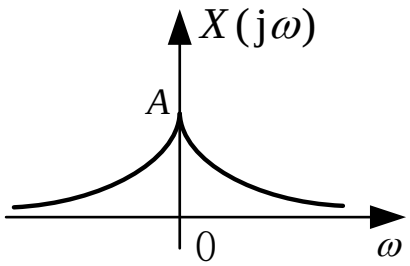
digital signal

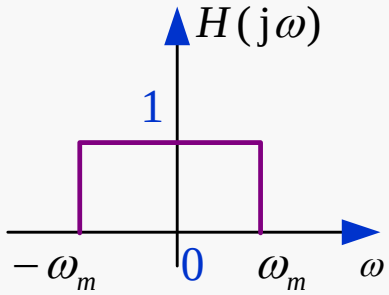


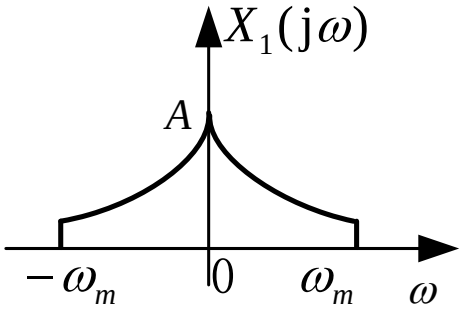
Sampling of non band limited signals

Many actual engineering signals do not meet the band limit conditions





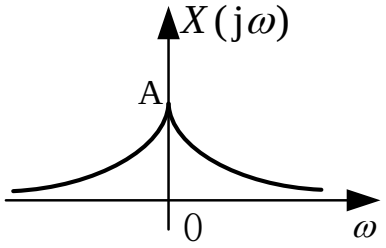


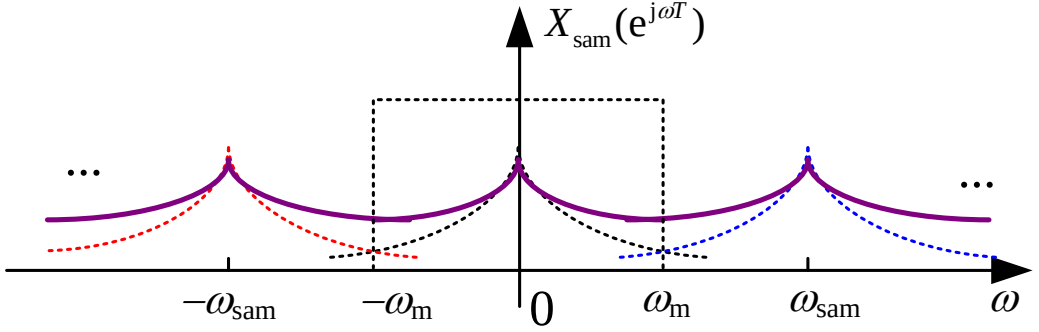


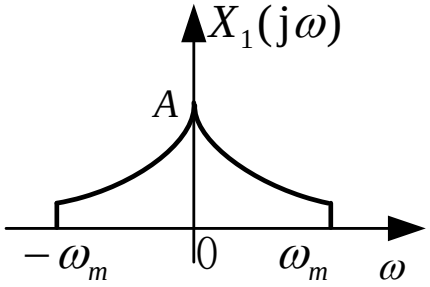
Sampling of non band limited signals

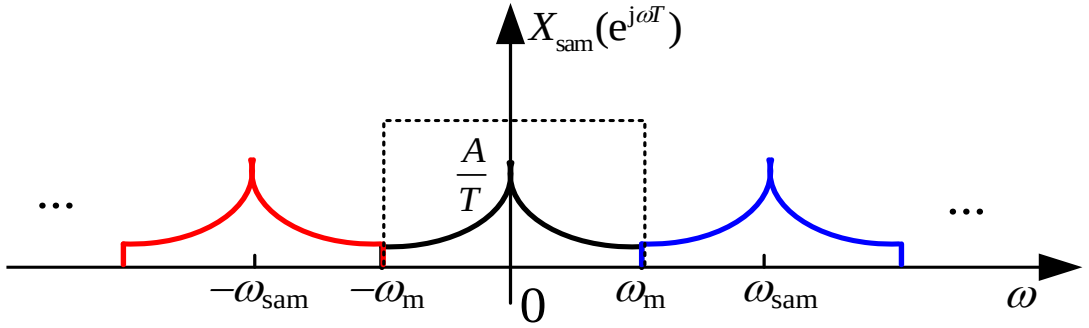
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Comparison of aliasing error and truncation error









Sampling of non band limited signals

Anti-aliasing filtering of speech signals

sampling frequency f_{sam}=44,100 Hz 

sampling frequency f_{sam}=5,512 Hz 

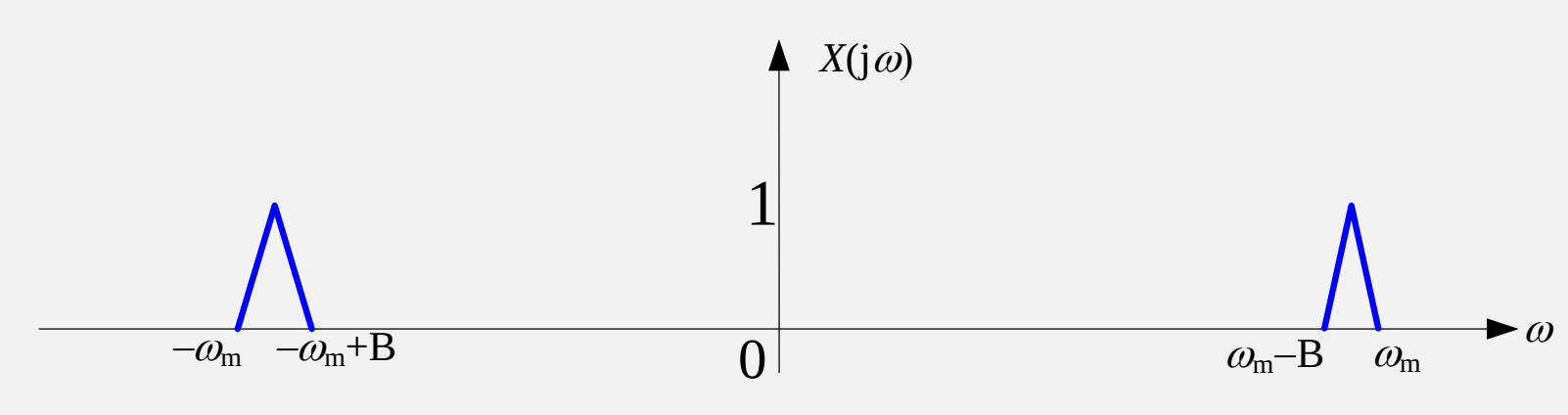
sampling frequency f_{sam}=5,512 Hz 

Anti aliasing filtering was applied to the signal before sampling

Sampling of narrowband high-frequency signals

Sampling problem of narrowband high-frequency signals

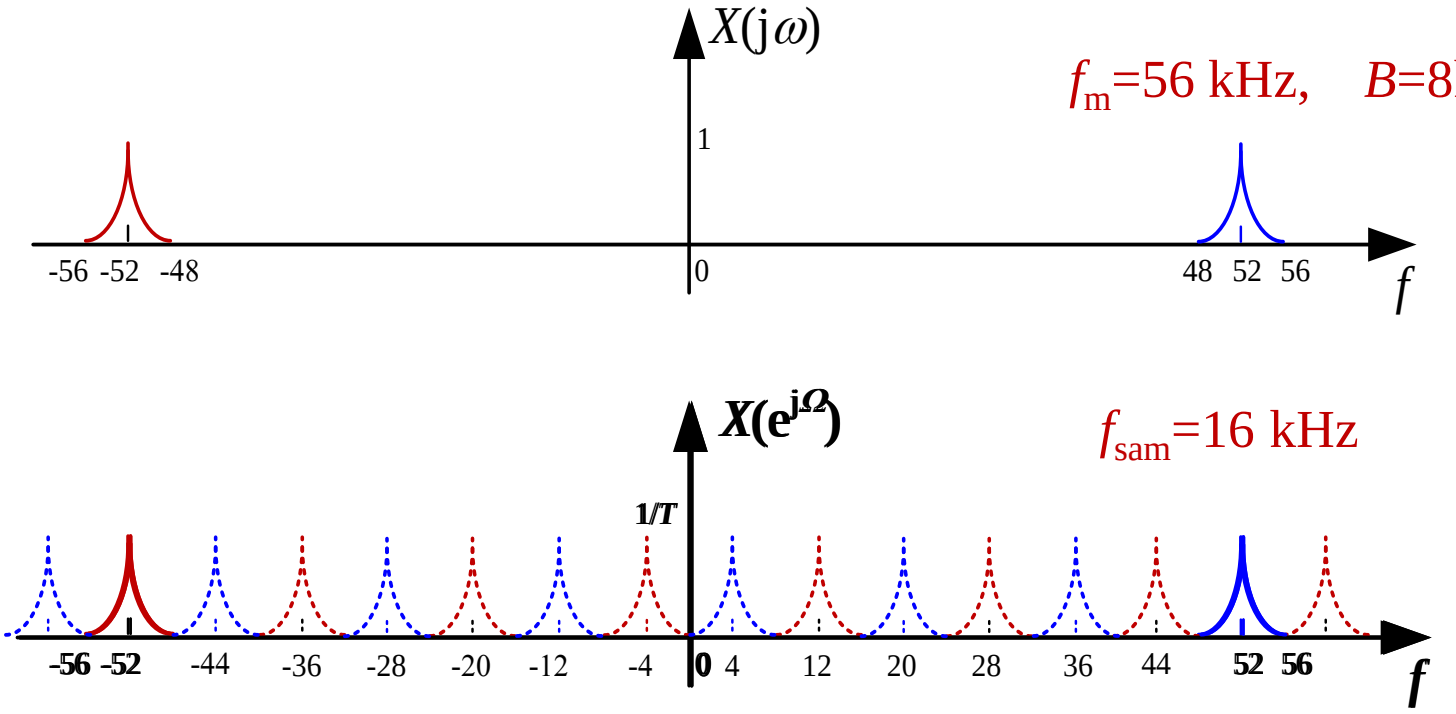
$$\omega_{sam} \geq 2B$$



$\omega_m = 10\text{M rad/s}, \quad B = 8\text{k rad/s}$

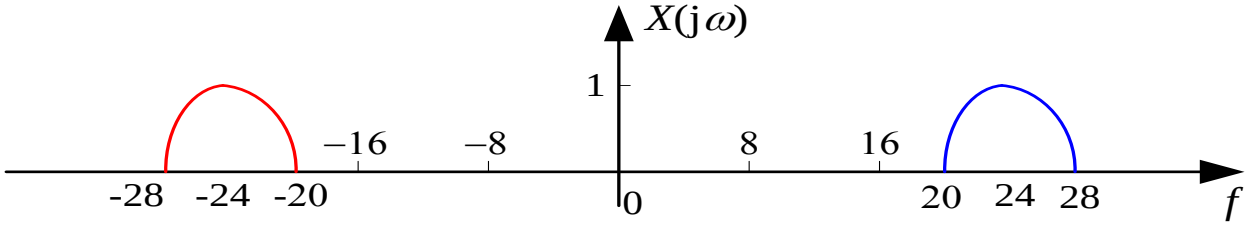
Sampling of narrowband high-frequency signals

Lancaster University College at Beijing Jiaotong University

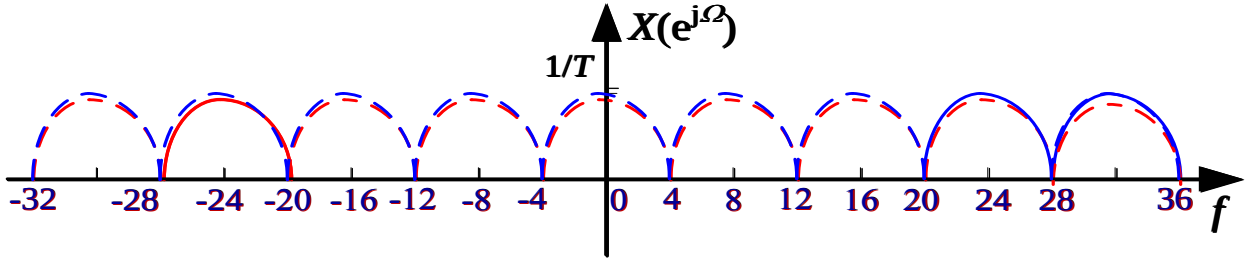


Sampling of narrowband high-frequency signals

Lancaster University College at Beijing Jiaotong University



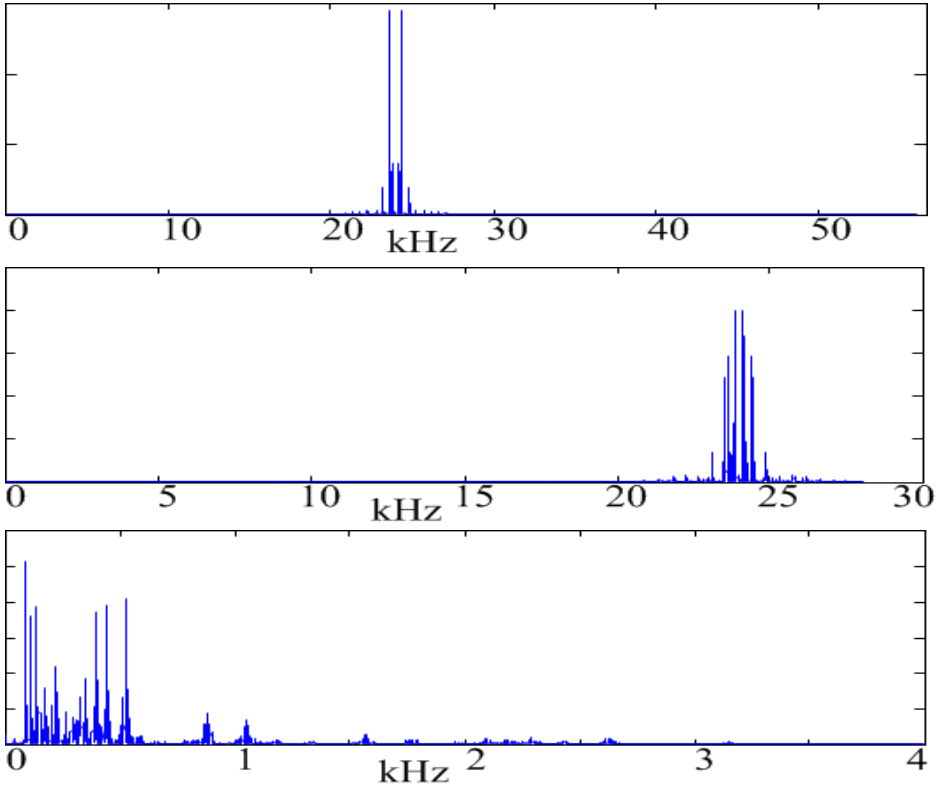
f_m=28 kHz $B = 8$k rad/s



f_{sam}=8 kHz

Sampling of narrowband high-frequency signals

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Sampling of narrowband high-frequency signals

The center frequency is 24kHz and the bandwidth is 8kHz.

Demodulated speech signal

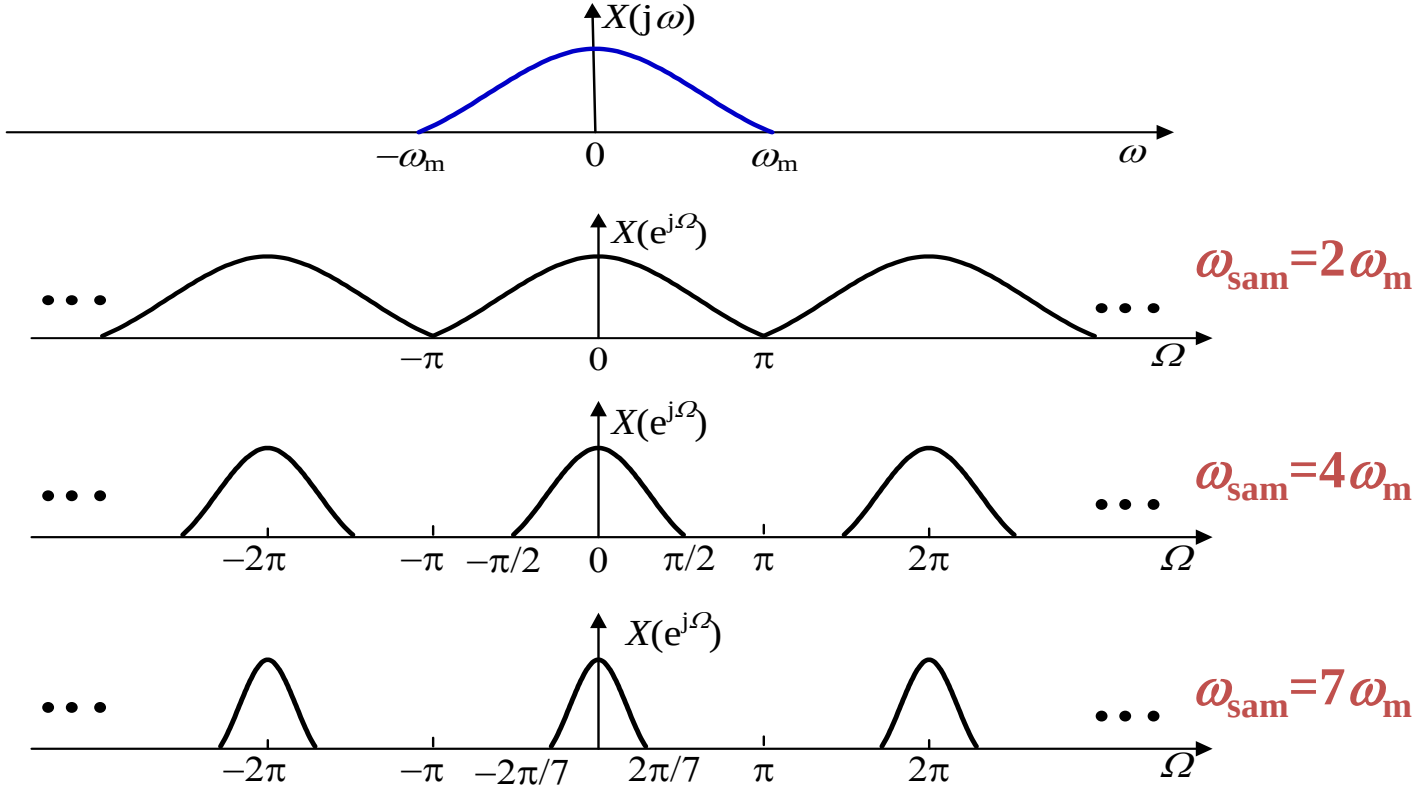
f_{sam}=56kHz the sampled spectrum.

Demodulated speech signal

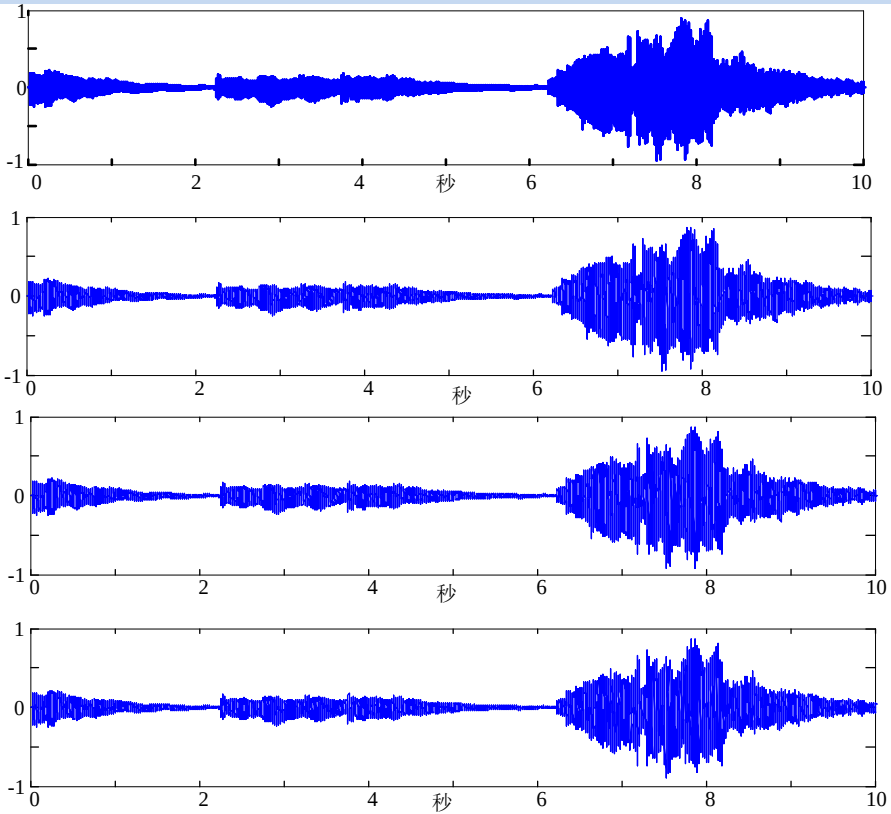
f_{sam}=8kHz the sampled spectrum.

Sampled speech signal (without demodulation)

Signal sampling at the highest frequency of an unknown signal



Signal sampling at the highest frequency of an unknown signal



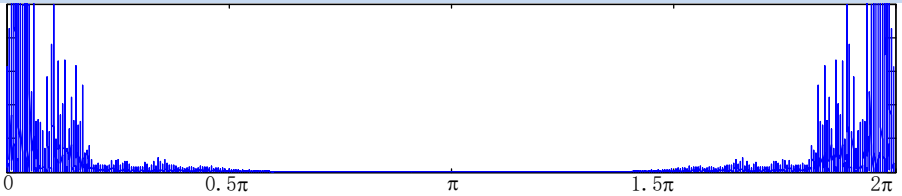
f_{sam}=44,100 Hz

f_{sam}=22,050 Hz

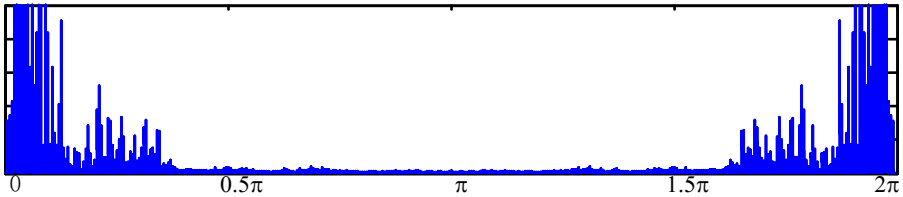
f_{sam}=11,025 Hz

f_{sam}=5,512 Hz

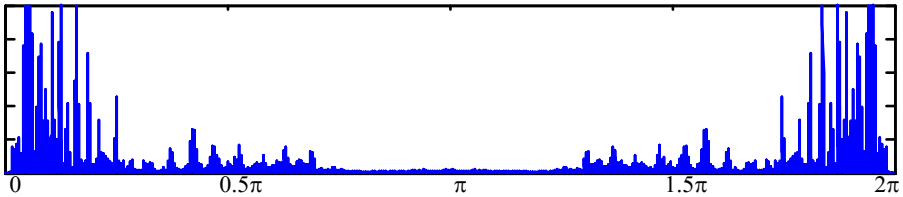
Signal sampling at the highest frequency of an unknown signal



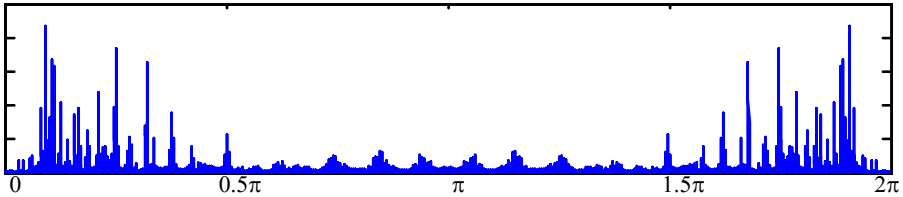
f_{sam}=44100 Hz



f_{sam}=22050 Hz



f_{sam}=11025 Hz



f_{sam}=5512 Hz

Railway control signal recognition

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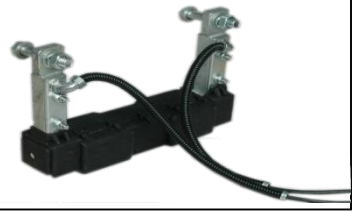
cab signal



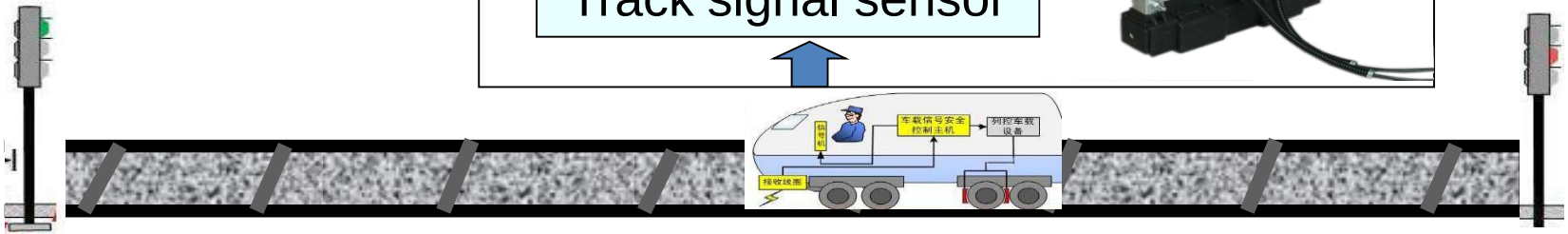
Cab signal recognition



Signal Sampling (A/D)



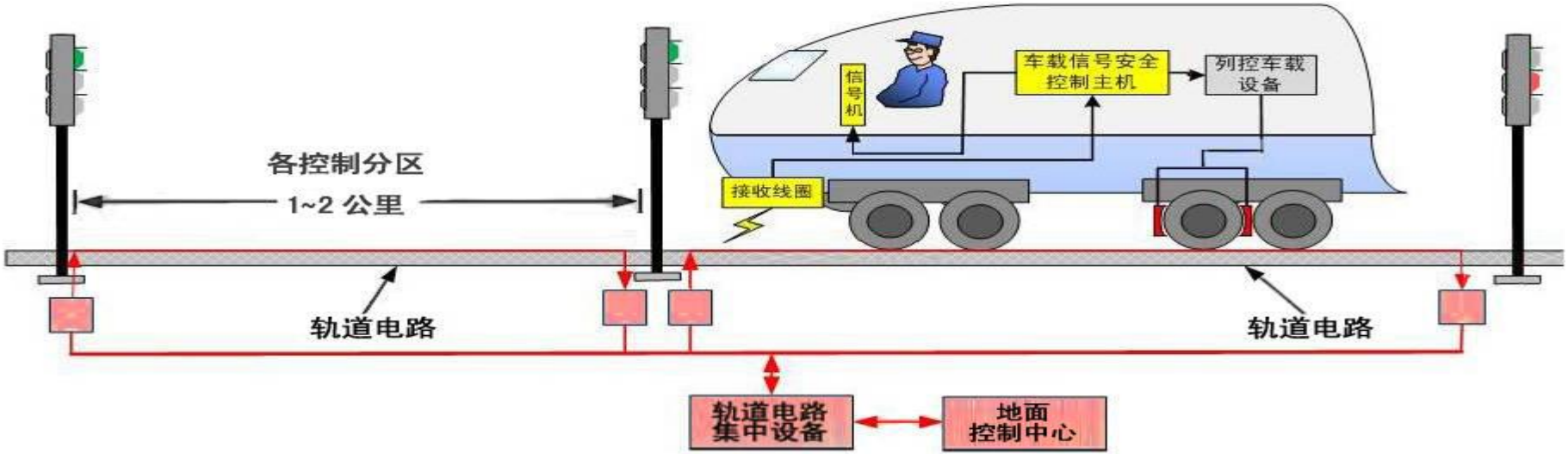
Track signal sensor



Railway control signal recognition

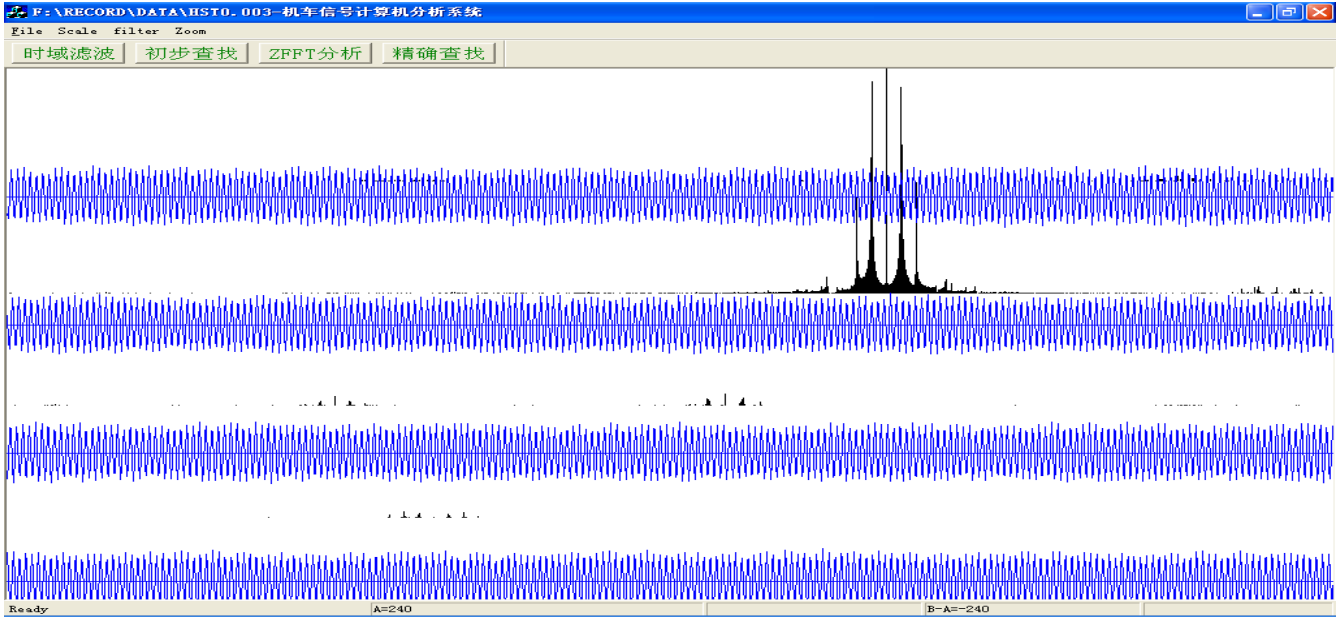
Lancaster University College at Beijing Jiaotong University

Traditional onboard signal systems, due to limitations in safety and reliability technologies, can only be used as auxiliary signals, and drivers must observe ground signals to drive trains. Internationally recognized, trains running at speeds above 160km/h or with high density can no longer rely on the driver's observation of ground signals to ensure safety, but must be controlled by onboard signals as the main signal.



Railway control signal recognition

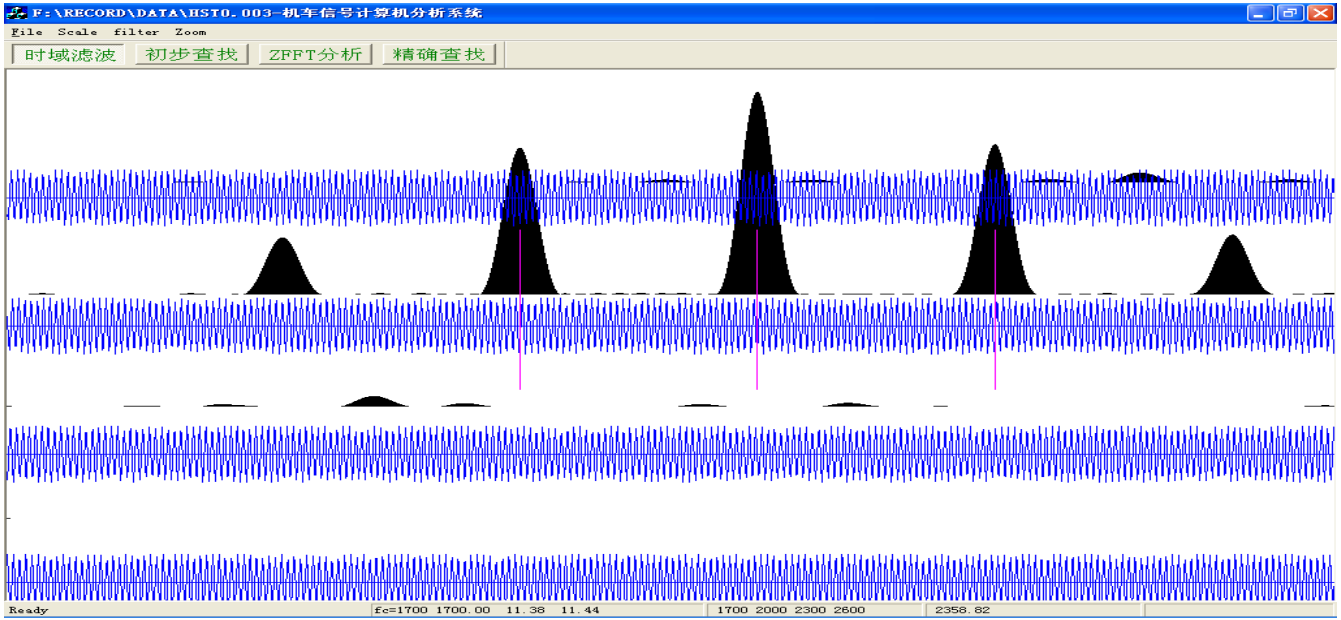
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Time domain waveform and spectrum of railway control signals

Railway control signal recognition

Lancaster University College at Beijing Jiaotong University

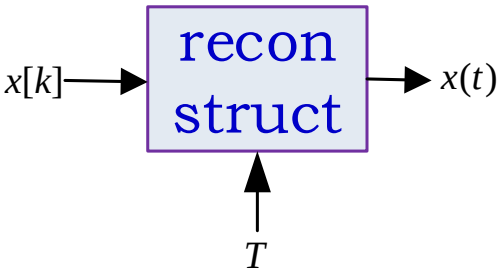


Spectrum analysis of railway control signals

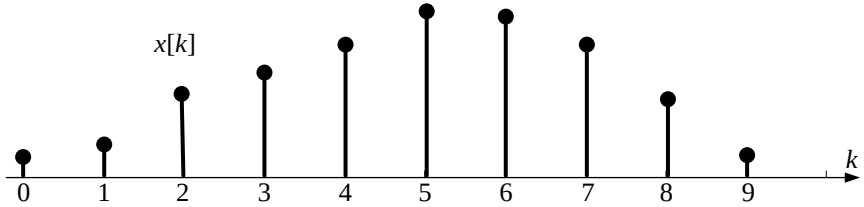
Reconstruction of sampled signals

- ※ Engineering Implementation of Signal Reconstruction
- ※ Theoretical analysis of signal reconstruction
- ※ Reconstruction interval for signal reconstruction

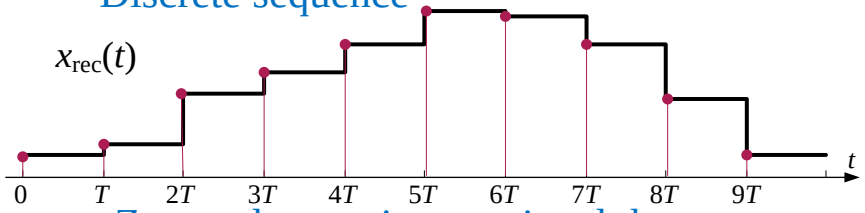
Reconstruction of sampled signals



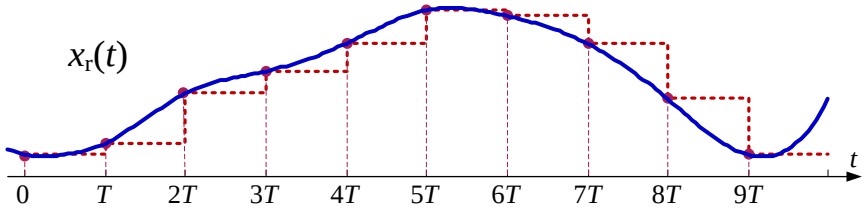




Discrete sequence



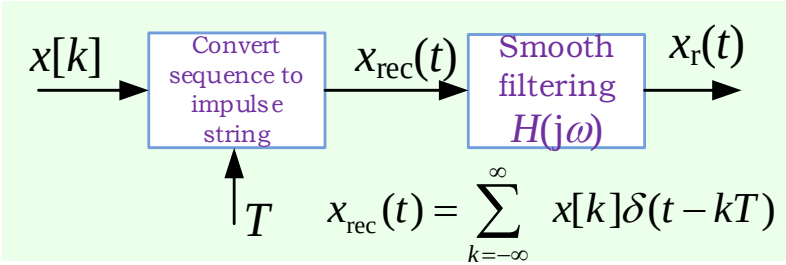
Zero order continuous signal that

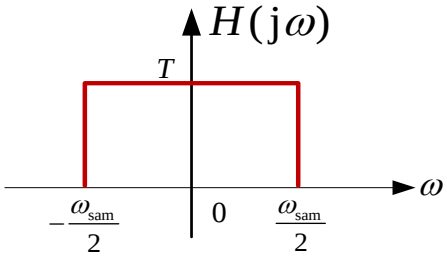


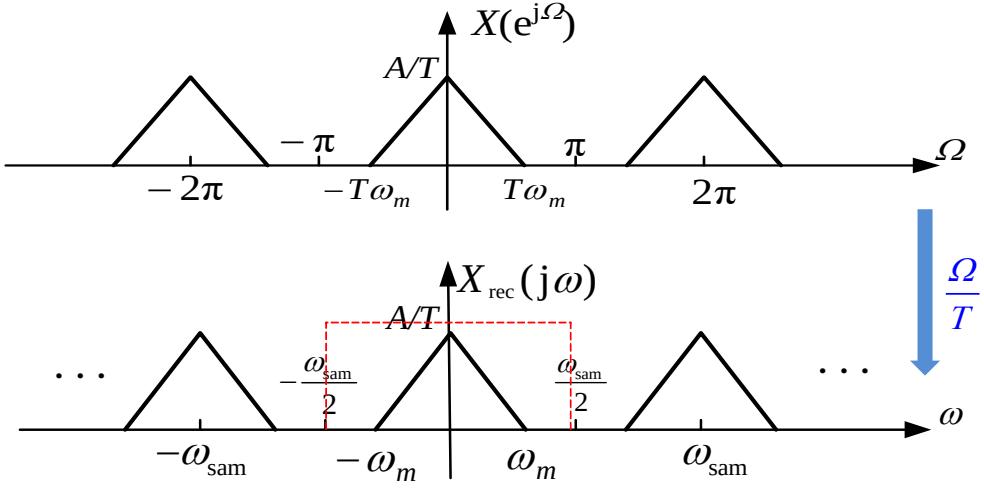
Reconstructed signal after low-pass filtering

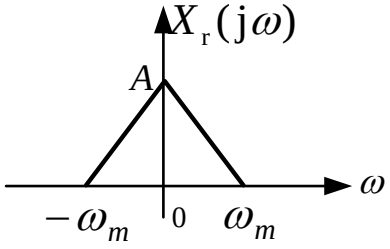
Reconstruction of sampled signals

※ Frequency
※ Domain



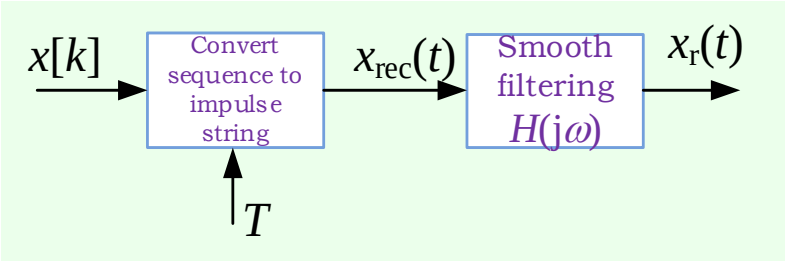


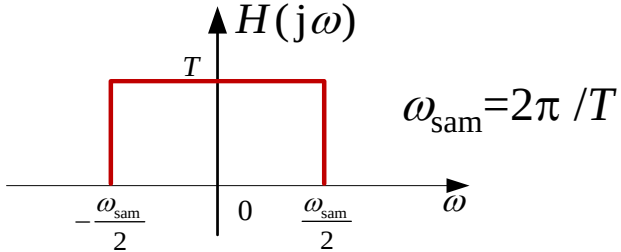




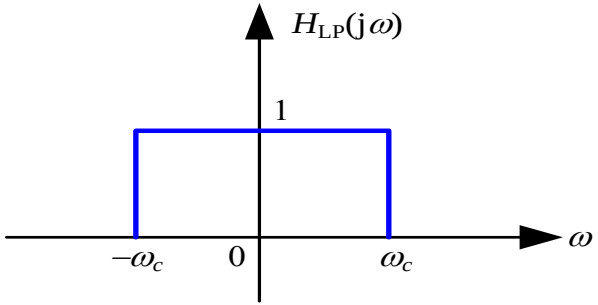
Reconstruction of sampled signals

※ Time Domain





$$h(t)=\frac{\omega_{sam}T}{2\pi}\mathrm{Sa}(\frac{\omega_{sam}}{2}t)=\mathrm{Sa}(\frac{\pi}{T}t)$$

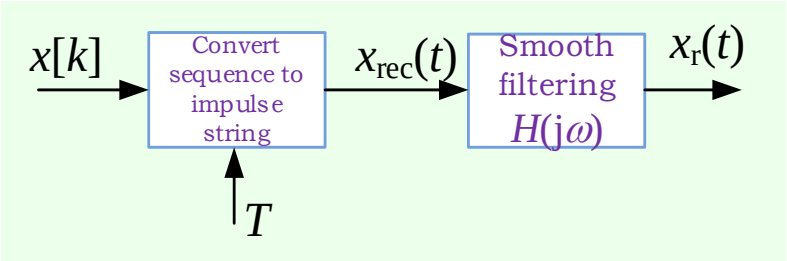


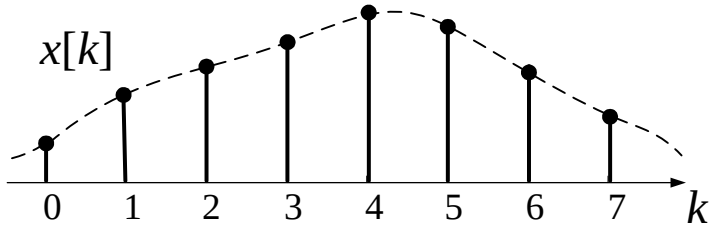
$$h(t)=\frac{\omega_c}{\pi}\mathrm{Sa}(\omega_c t)$$

Reconstruction of sampled signals

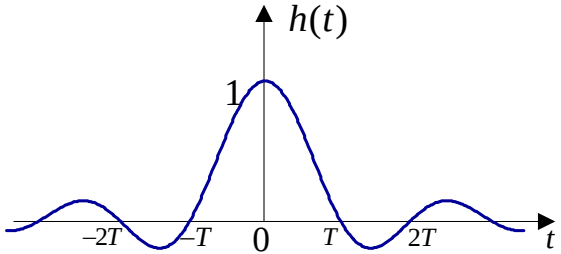
※ Time Domain

$$x_{rec}(t) = \sum_{k=-\infty}^{\infty} x[k]\delta(t-kT)$$



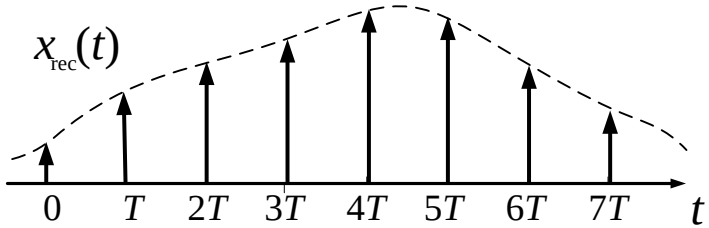


Discrete sequence



Filter impulse response

$$h(t) = \mathrm{Sa}(\frac{\pi}{T}t)$$

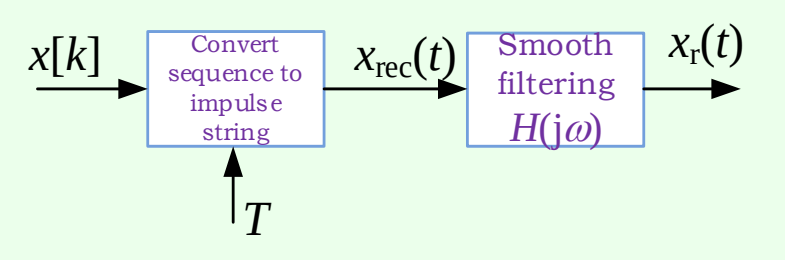


Impulse string signal

Reconstruction of sampled signals

※ Time Domain

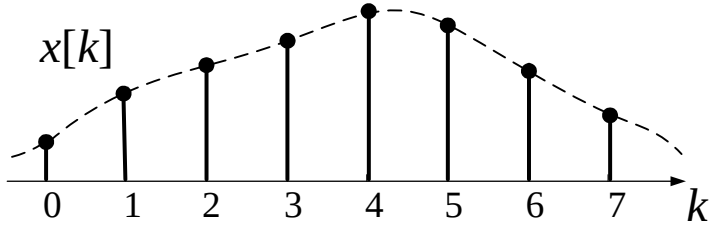
$$x_{rec}(t)=\sum_{k=-\infty}^{\infty}x[k]\delta(t-kT)$$



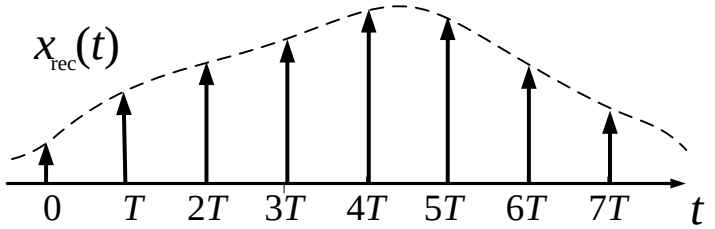
$$x_r(t)=x_{rec}(t)*h(t)$$

$$=\sum_{k=-\infty}^{\infty}x[k]\delta(t-kT)*\mathrm{Sa}(\frac{\pi}{T}t)$$

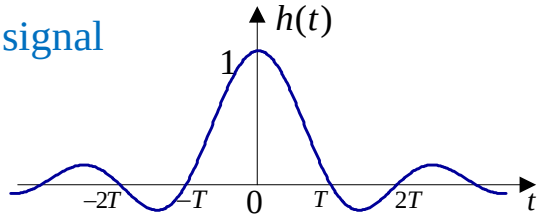
$$=\sum_{k}x[k]\mathrm{Sa}(\frac{\pi}{T}(t-kT))$$



Discrete sequence



Impulse string signal



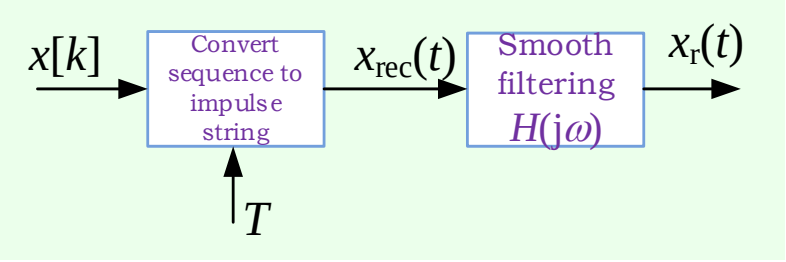
Filter impulse response

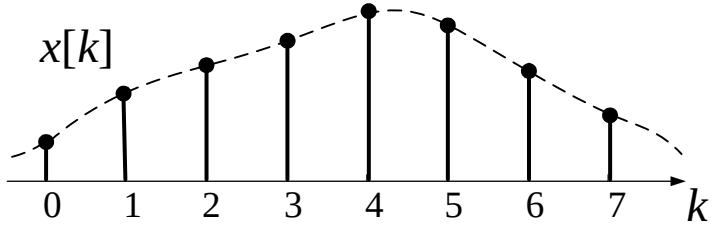
Reconstruction of sampled signals

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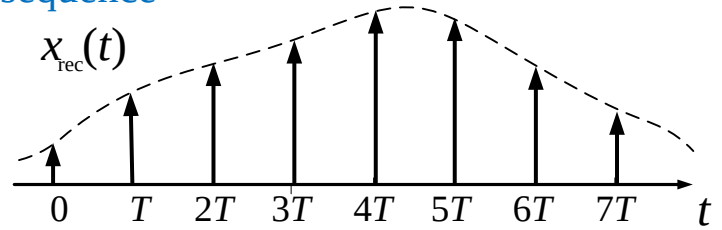
※ Time Domain

$$x_{rec}(t)=\sum_{k=-\infty}^{\infty} x[k]\delta(t-kT)$$



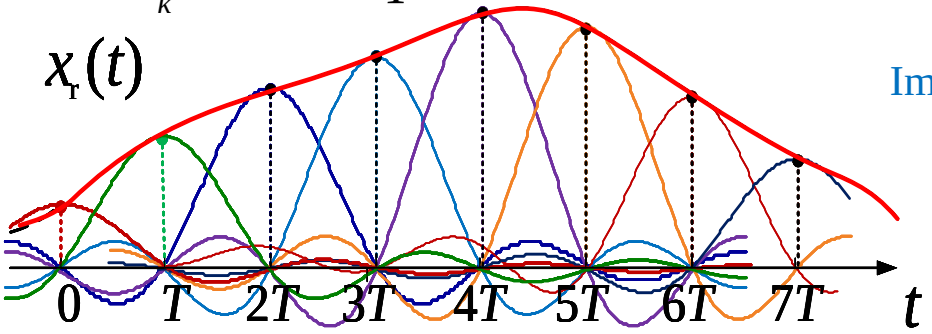


Discrete sequence

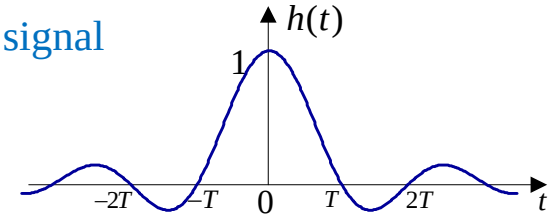


Impulse string signal

$$x_r(t)=\sum_k x[k]\mathrm{Sa}(\frac{\pi}{T}(t-kT))$$



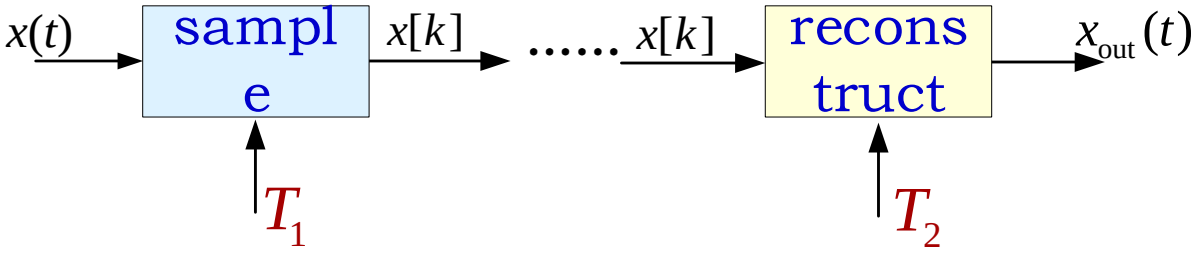
Reconstructed continuous signal



Filter impulse response

Reconstruction interval for signal reconstruction

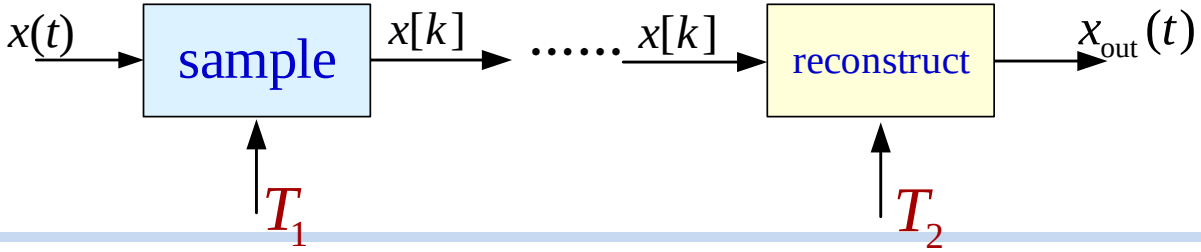
- The relationship between sampling interval and reconstruction interval

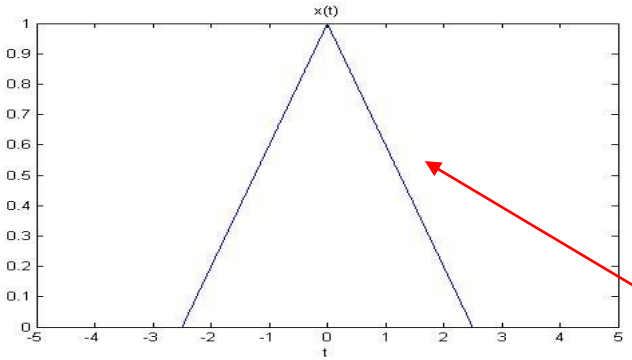


In order to ensure the correctness of the reconstructed signal, it is required

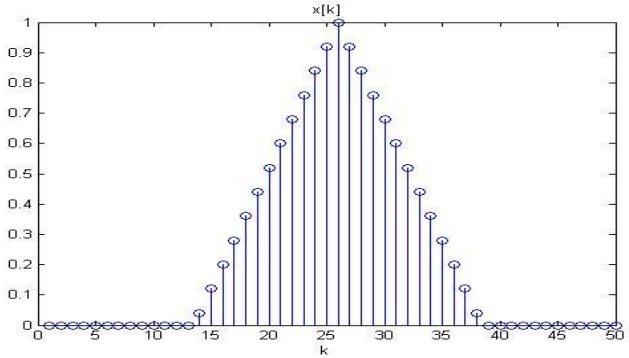
$$T_1 = T_2$$

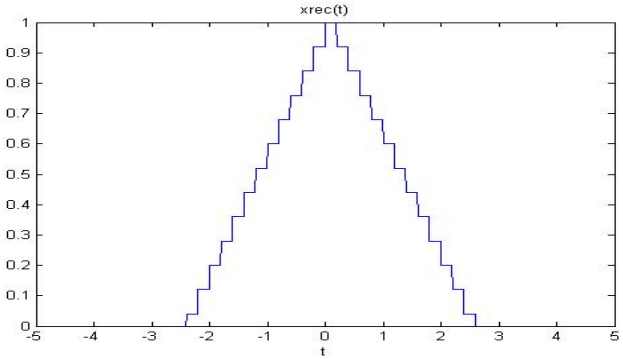
What problems will occur if T1 ≠ T2?



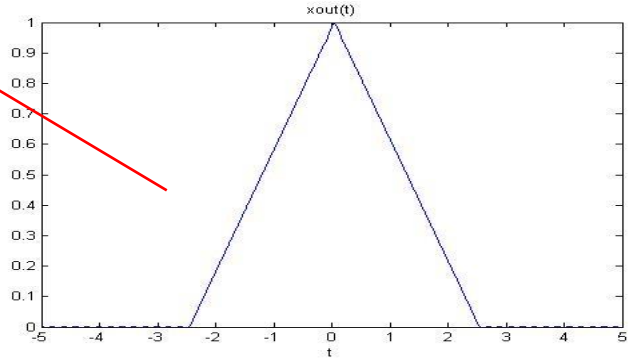


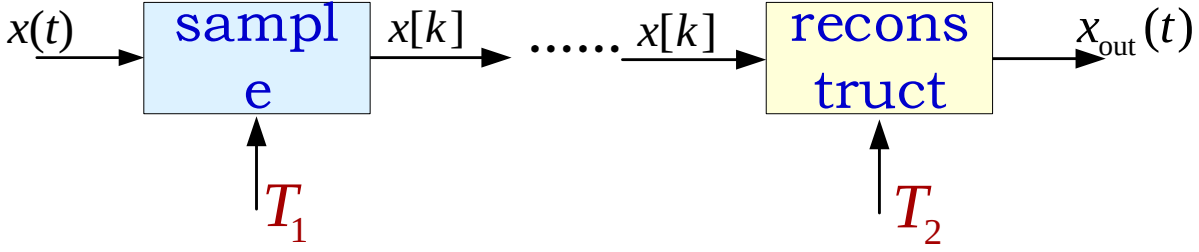
$$T_2 = T_1$$

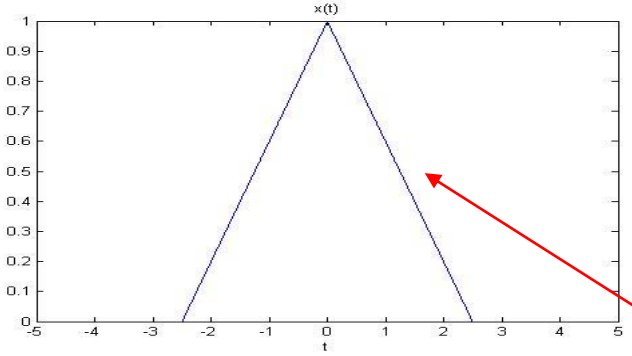




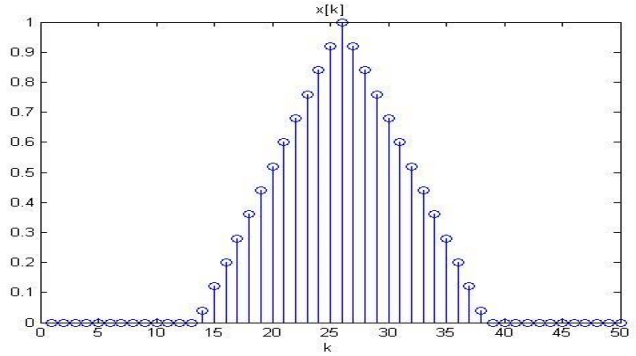
$$x_{out}(t) \approx x(t)$$

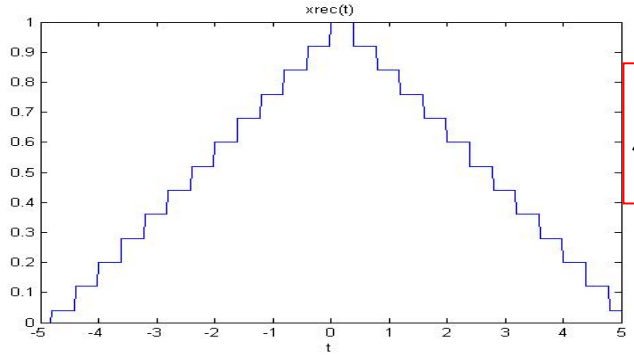




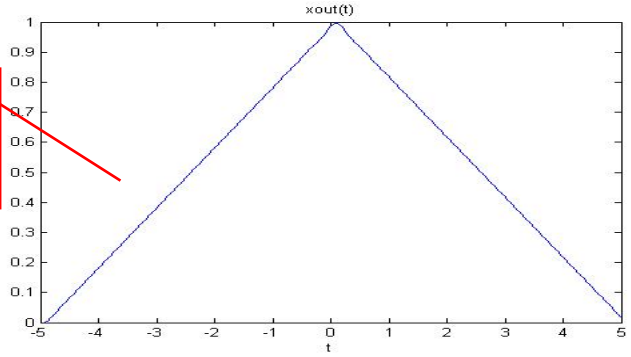


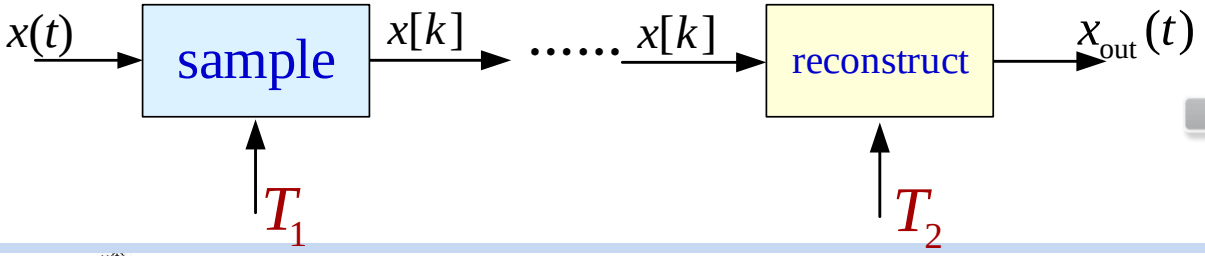
$$T_2 = 2T_1$$

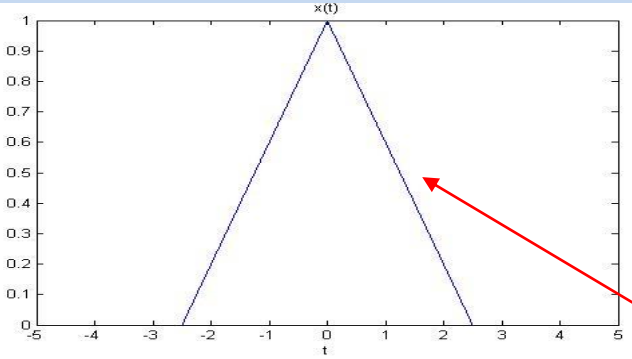




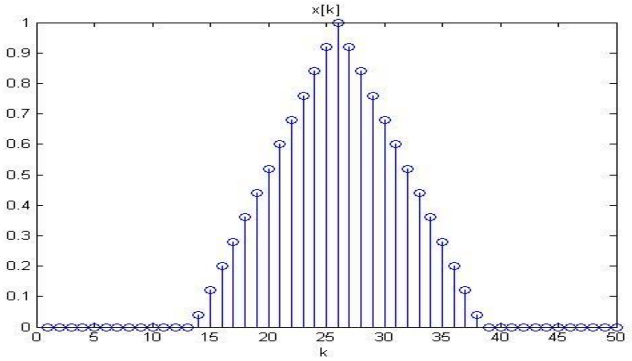
$$x_{out}(t) \approx x(\frac{t}{2})$$

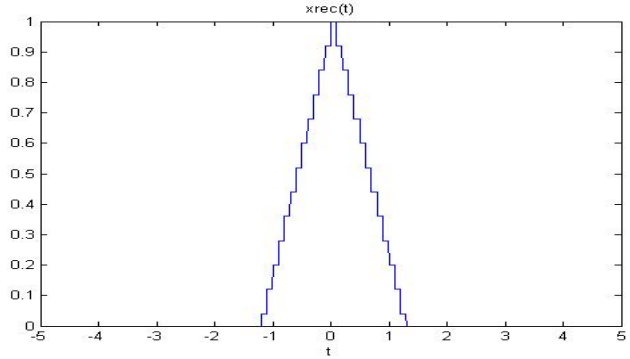




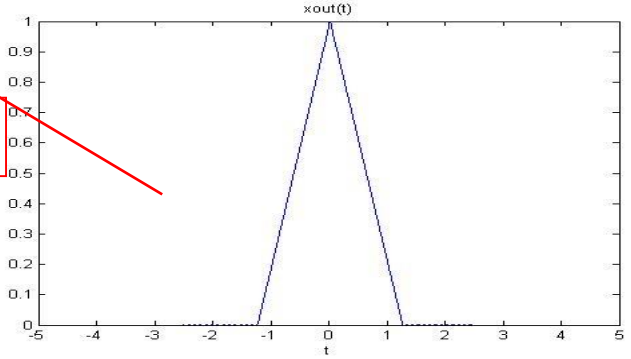


$$T_2 = T_1/2$$

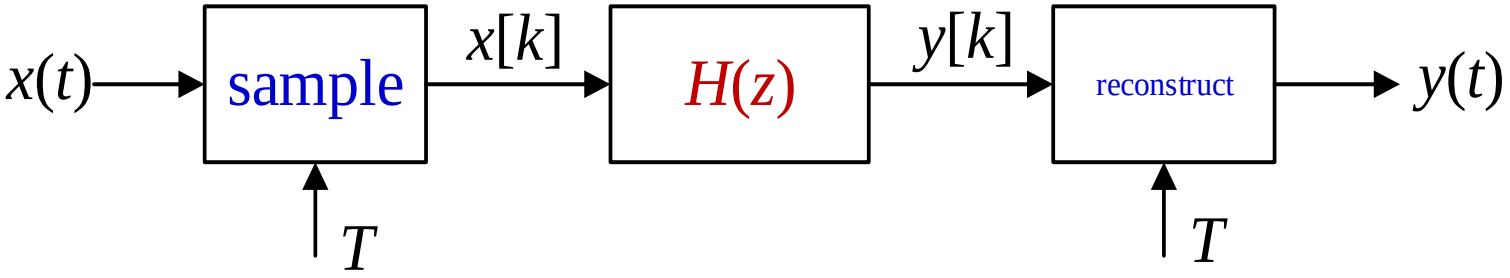




$$x_{out}(t) \approx x(2t)$$

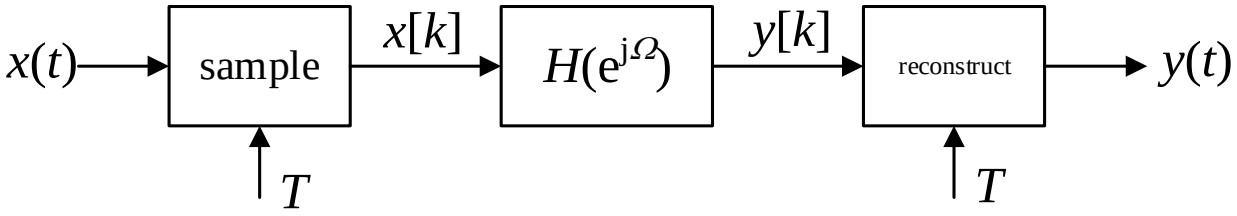


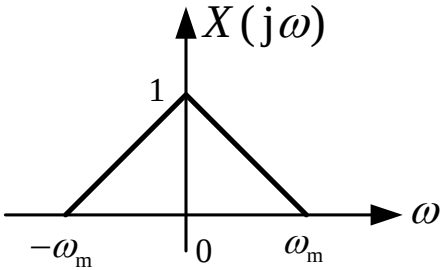
Discrete processing of continuous signals



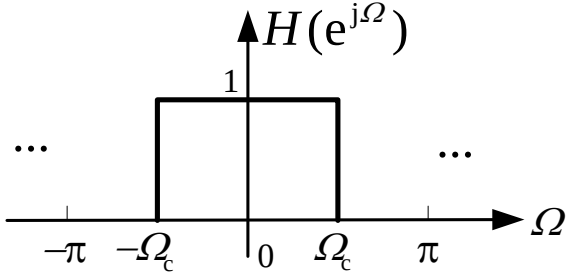
Discrete processing of continuous signals

E.g: As shown in the system diagram, given the frequency spectrum $X(\mathrm{j}w)$ of the input signal $x(t)$ and the frequency response $H(\mathrm{e}^{\mathrm{j}\Omega})$ of the discrete system, plot the spectra $X(\mathrm{e}^{\mathrm{j}\Omega})$, $Y(\mathrm{e}^{\mathrm{j}\Omega})$, and $Y(\mathrm{j}\omega)$ of $x[k]$, $y[k]$, and $y(t)$.

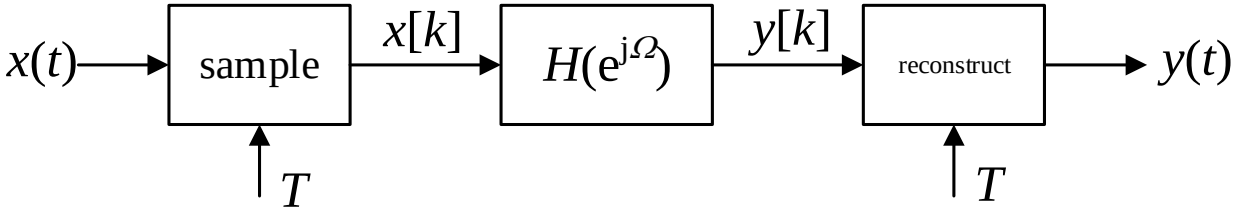




$$\omega_{\text{sam}} = 2\omega_{\text{m}}$$

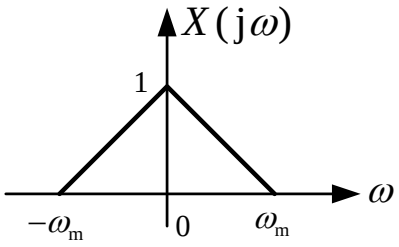


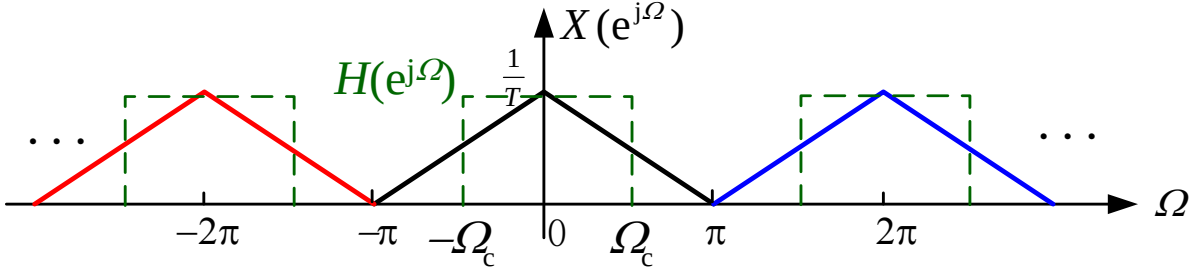
Discrete processing of continuous signals



Solution:

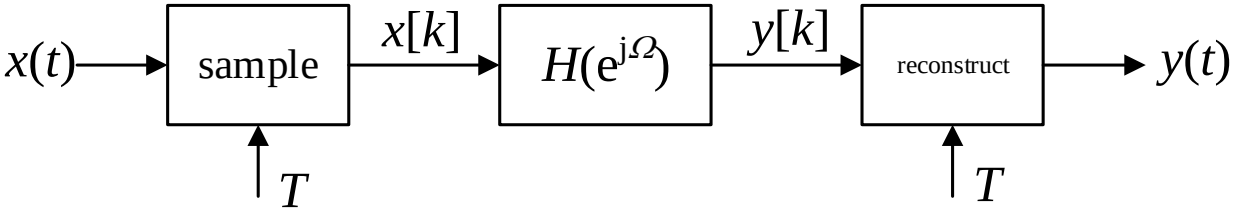
$$X(e^{j\Omega}) = \frac{1}{T}\sum_{n=-\infty}^{+\infty} X\left[j(\omega - n\omega_{sam})\right] \qquad (\Omega = \omega T)$$



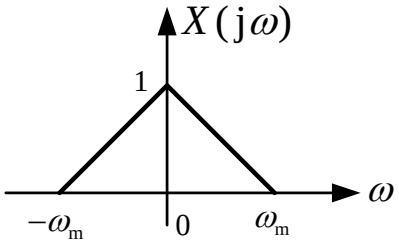


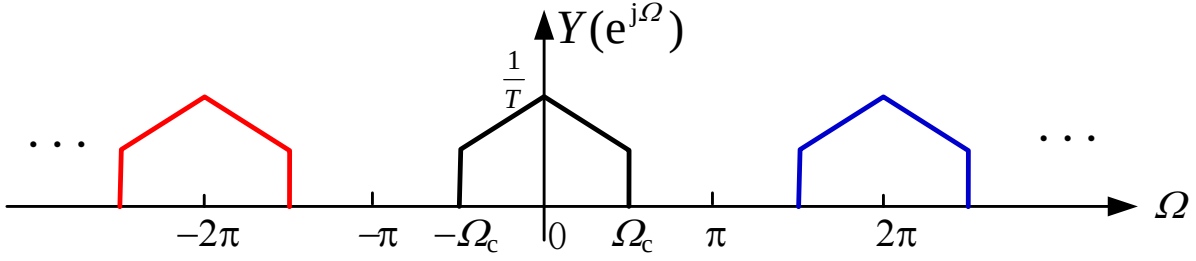
$$Y(e^{j\Omega}) = X(e^{j\Omega})H(e^{j\Omega})$$

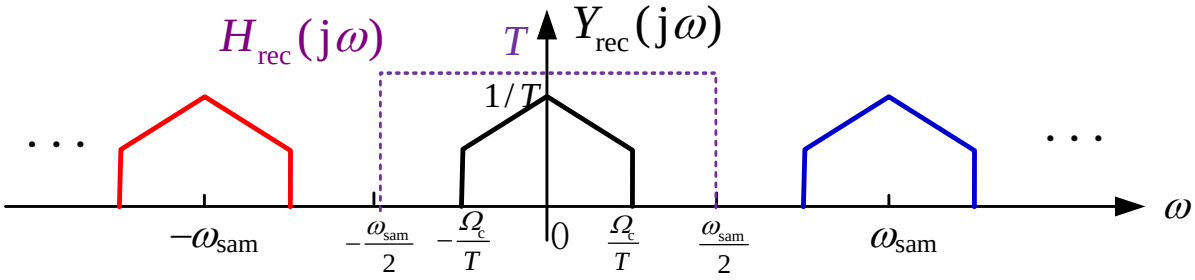
Discrete processing of continuous signals



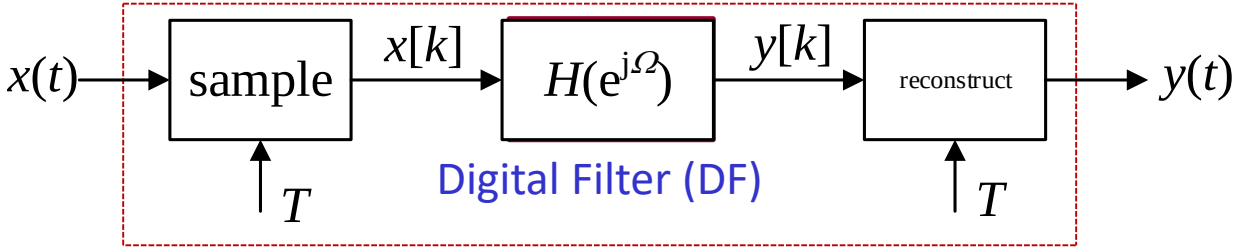
Solution:





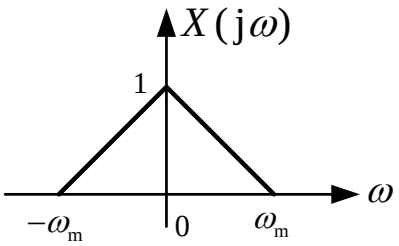


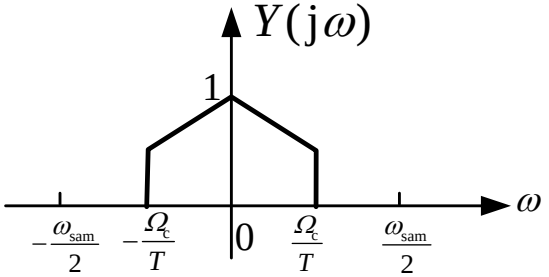
Discrete processing of continuous signals



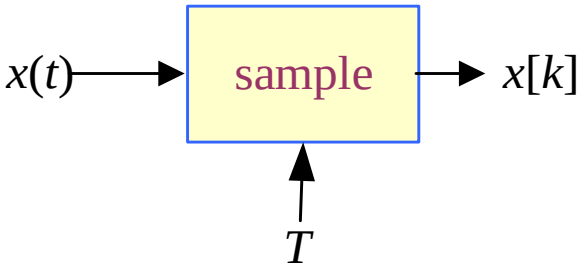
Solution:

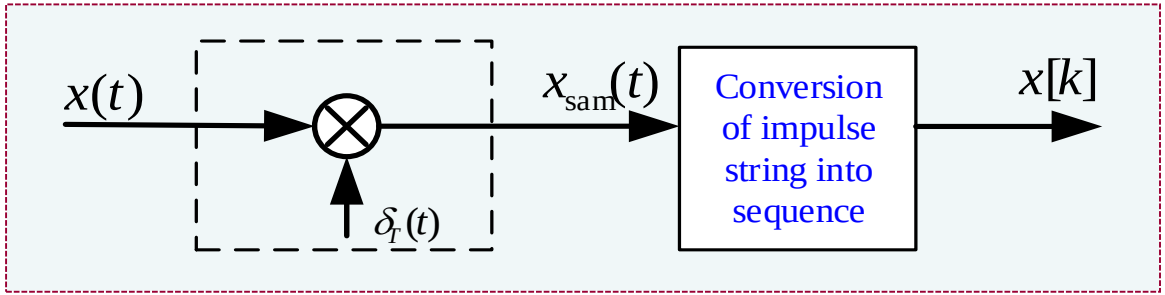
Equivalent Analog Filter (AF)

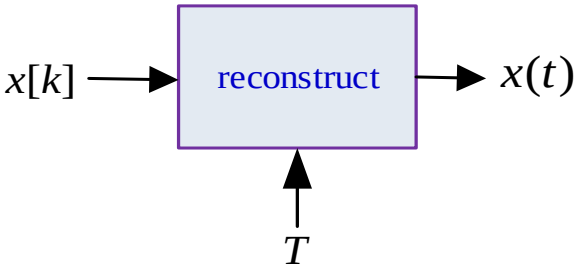


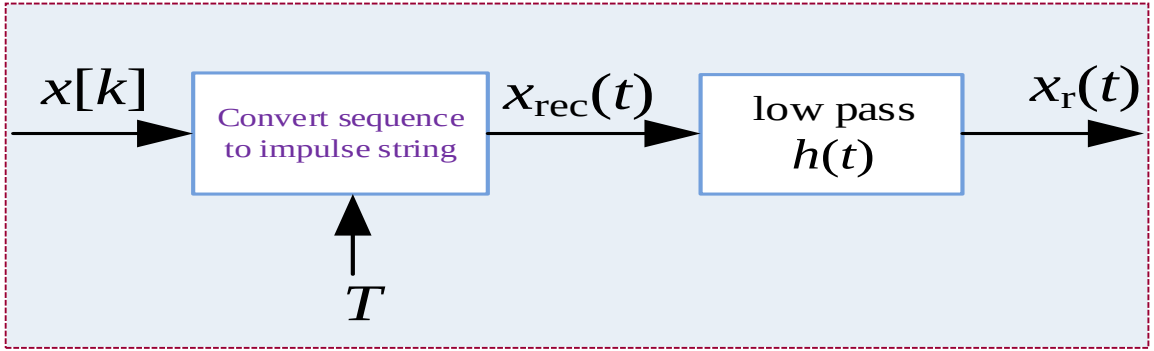


Discrete processing of continuous signals



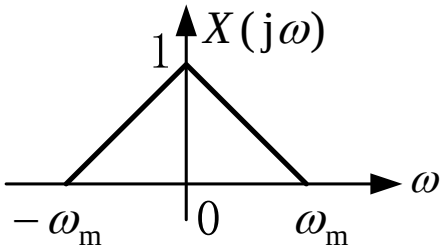


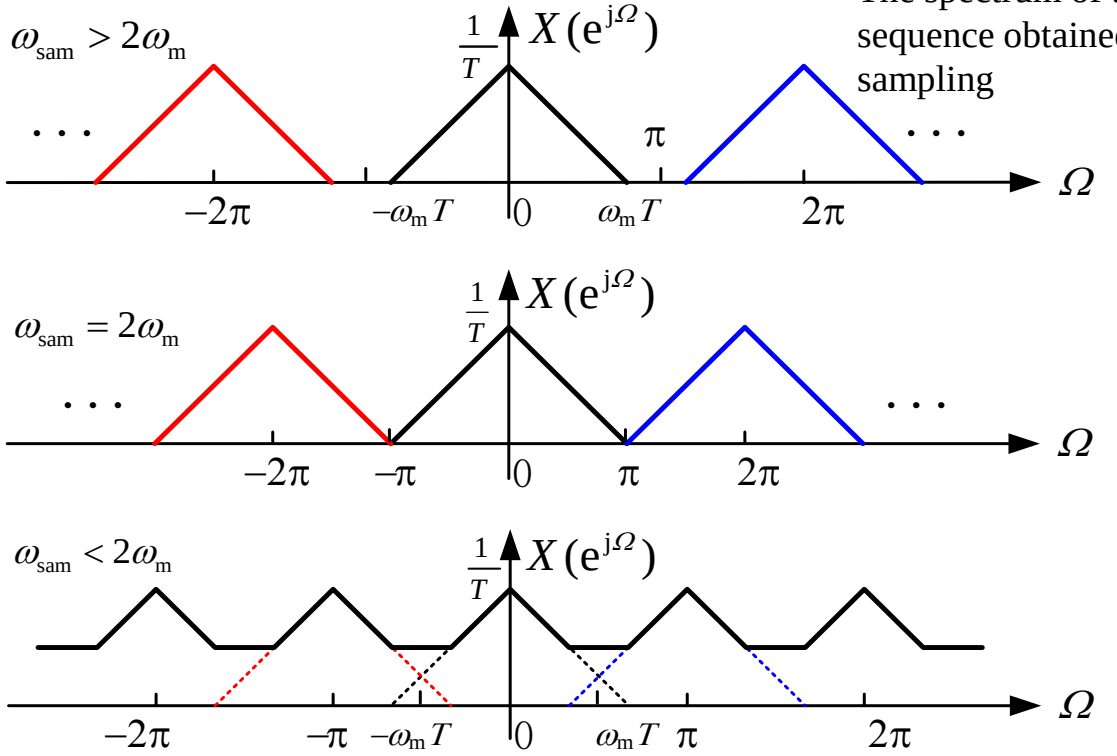




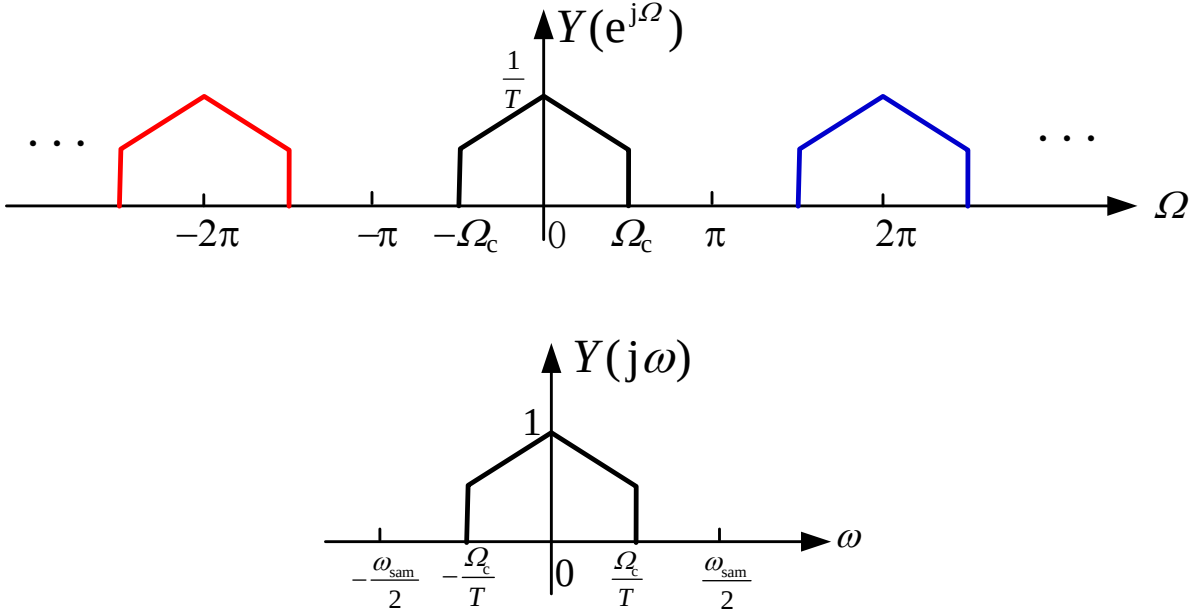
$$X(\mathrm{e}^{\mathrm{j}\Omega}) = \frac{1}{T}\sum_{n=-\infty}^{+\infty} X\left[\mathrm{j}(\omega - n\omega_{\mathrm{sam}})\right] \qquad (\Omega = \omega T)$$

The spectrum of the discrete sequence obtained from sampling

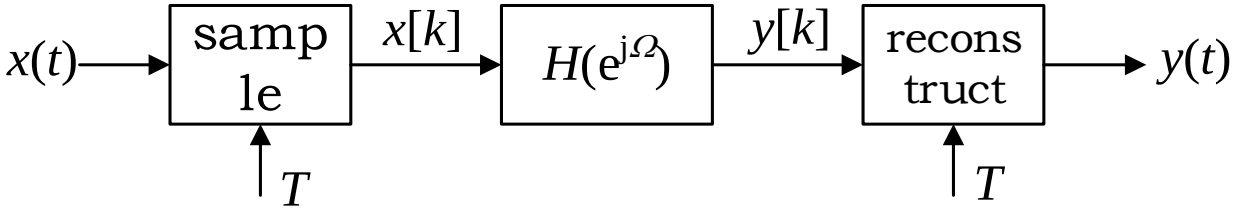


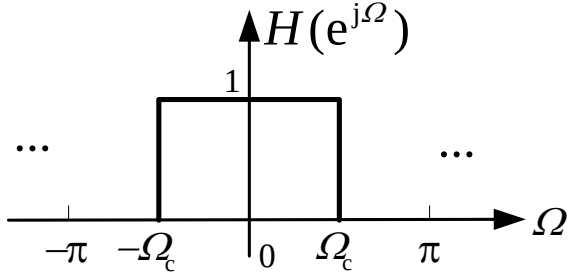


Time domain reconstruction



Discrete processing of continuous signals

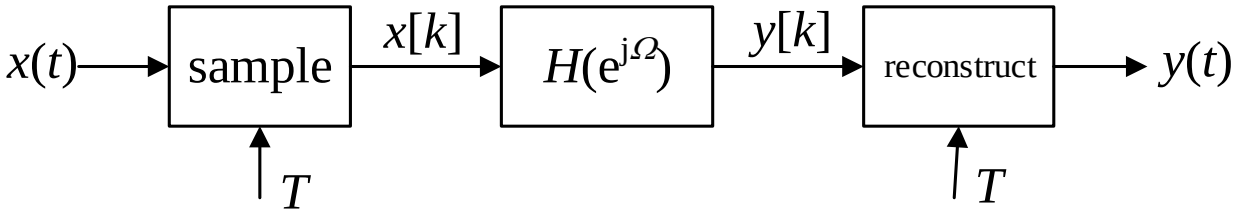


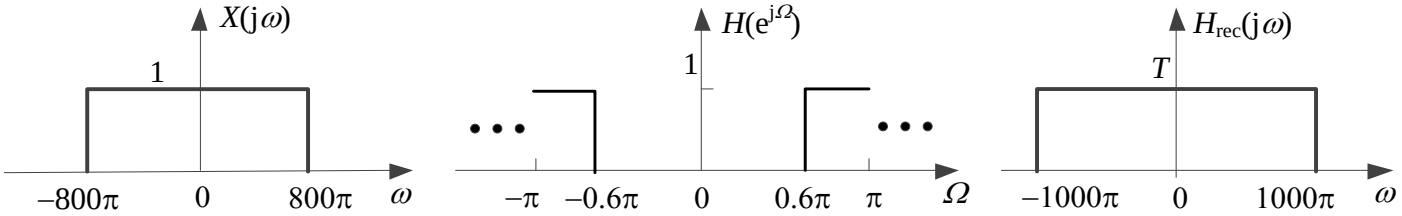


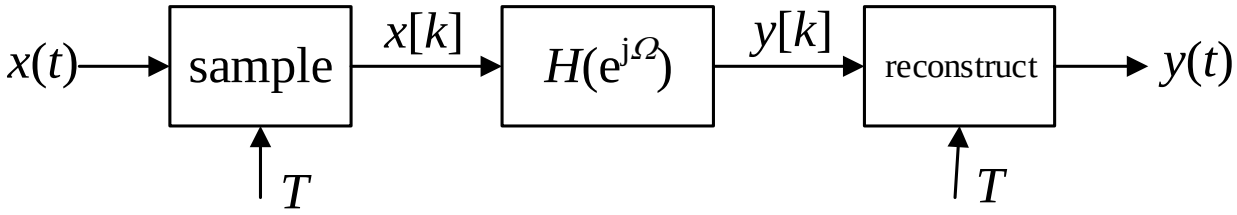
As shown in the block diagram for processing analog signals using a digital system, the sampling interval of the system is T=1/1000 seconds.

(1) Write the relationship between $x[k]$ spectrum and $x(t)$ spectrum, $x[k]$ spectrum and $w[k]$ spectrum, and $w[k]$ spectrum and $y[k]$ spectrum.

(2)Draw the frequency spectra of signals$x[k]$, $w[k]$, $y[k]$, $y(t)$ separately.







(1)Write the relationship between $x[k]$ spectrum and $x(t)$ spectrum, $x[k]$ spectrum and $w[k]$ spectrum, and $w[k]$ spectrum and $y[k]$ spectrum.

The relationship between $x[k]$ spectrum and $x(t)$ spectrum

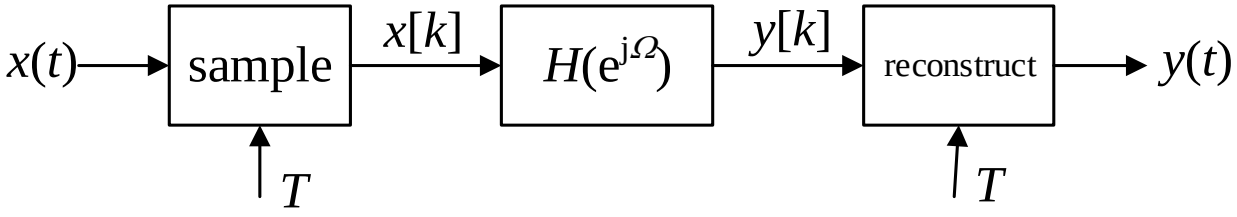
$$X(\mathrm{e}^{\mathrm{j}\Omega}) = \frac{1}{T}\sum_{n=-\infty}^{\infty} X[\mathrm{j}(\omega - n\omega_{\mathrm{sam}})] \qquad (\omega_{\mathrm{sam}}=2\pi/T,\ \Omega=\omega T)$$

The relationship between $x[k]$ spectrum and $w[k]$ spectrum

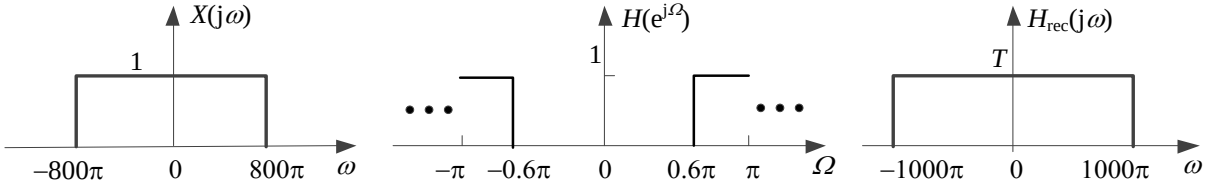
$$w[k] = x[k](-1)^k = x[k]\mathrm{e}^{\mathrm{j}\pi k} \qquad W(\mathrm{e}^{\mathrm{j}\Omega}) = X(\mathrm{e}^{\mathrm{j}(\Omega-\pi)})$$

The relationship between $w[k]$ spectrum and $y[k]$ spectrum

$$Y(\mathrm{e}^{\mathrm{j}\Omega}) = W(\mathrm{e}^{\mathrm{j}\Omega})H(\mathrm{e}^{\mathrm{j}\Omega})$$

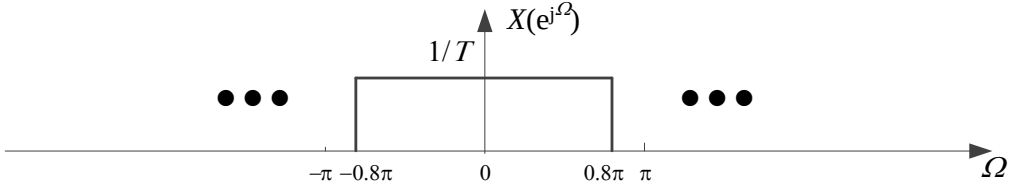


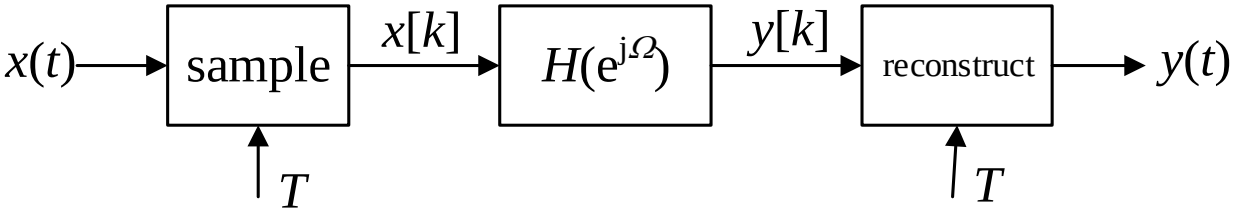
(2)Draw the frequency spectra of signals$x[k]$, $w[k]$, $y[k]$, $y(t)$ separately.



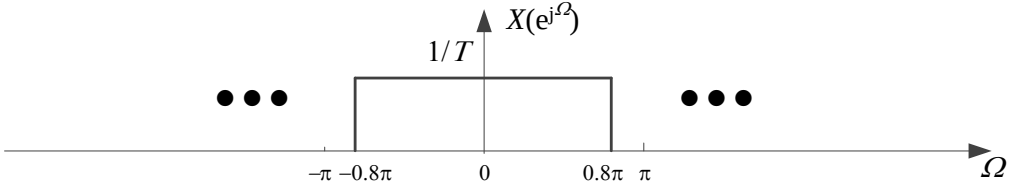
Spectrum of $x[k]$

$$X(\mathrm{e}^{\mathrm{j}\Omega}) = \frac{1}{T}\sum_{n=-\infty}^{\infty} X[\mathrm{j}(\omega - n\omega_{\mathrm{sam}})] \qquad (\omega_{\mathrm{sam}}=2\pi/T,\ \Omega=\omega T)$$



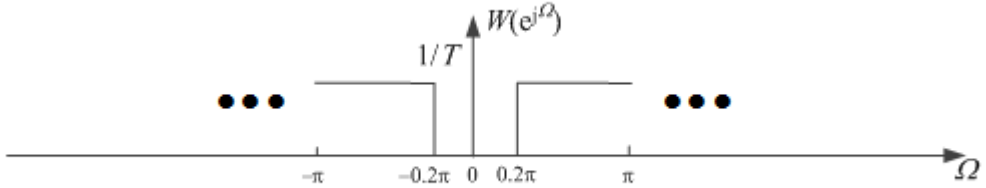


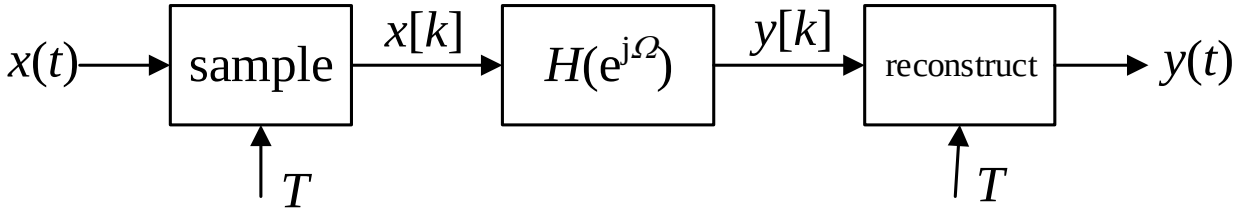
(2)Draw the frequency spectra of signals$x[k]$, $w[k]$, $y[k]$, $y(t)$ separately.



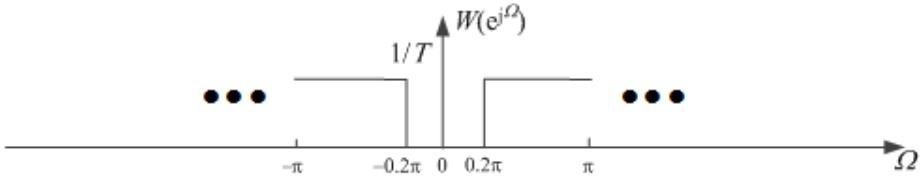
Spectrum of $w[k]$

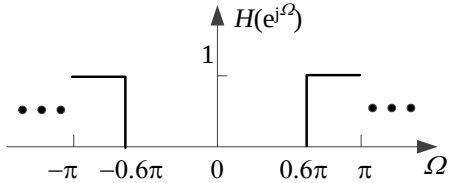
$$W(e^{j\Omega}) = X(e^{j(\Omega-\pi)})$$





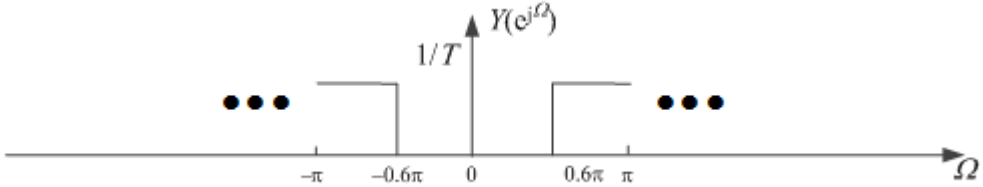
(2)Draw the frequency spectra of signals$x[k]$, $w[k]$, $y[k]$, $y(t)$ separately.

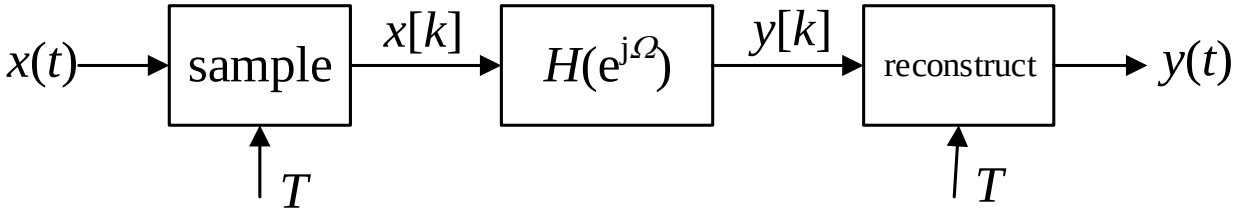




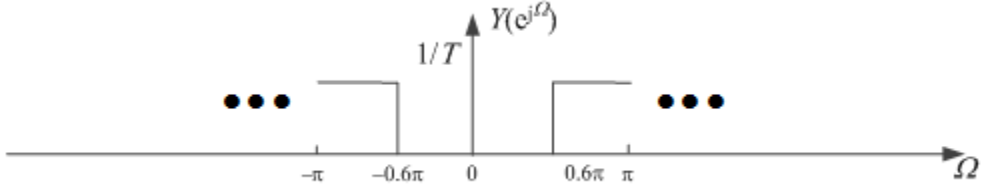
Spectrum of $y[k]$

$$Y(e^{j\Omega}) = W(e^{j\Omega})H(e^{j\Omega})$$

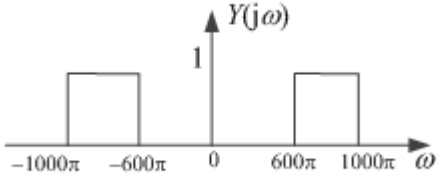




(2)Draw the frequency spectra of signals$x[k]$, $w[k]$, $y[k]$, $y(t)$ separately.



Spectrum of $y(t)$



Thanks

The materials referenced in this course are accumulated over years of teaching by the instructor, sourced from various media and exchanges with colleagues and peers. It is difficult to indicate the source of each material one by one. This is to clarify and express gratitude.